



Psychology of E-Learning

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- **E-Learning Definition-**

- E-learning is any form of teaching, training or tutoring designed to meet the needs of identified learners of any age and ability by scheduled or continual provision via the internet or mobile telephones, using electronic multimedia resources, computers and computer based devices.

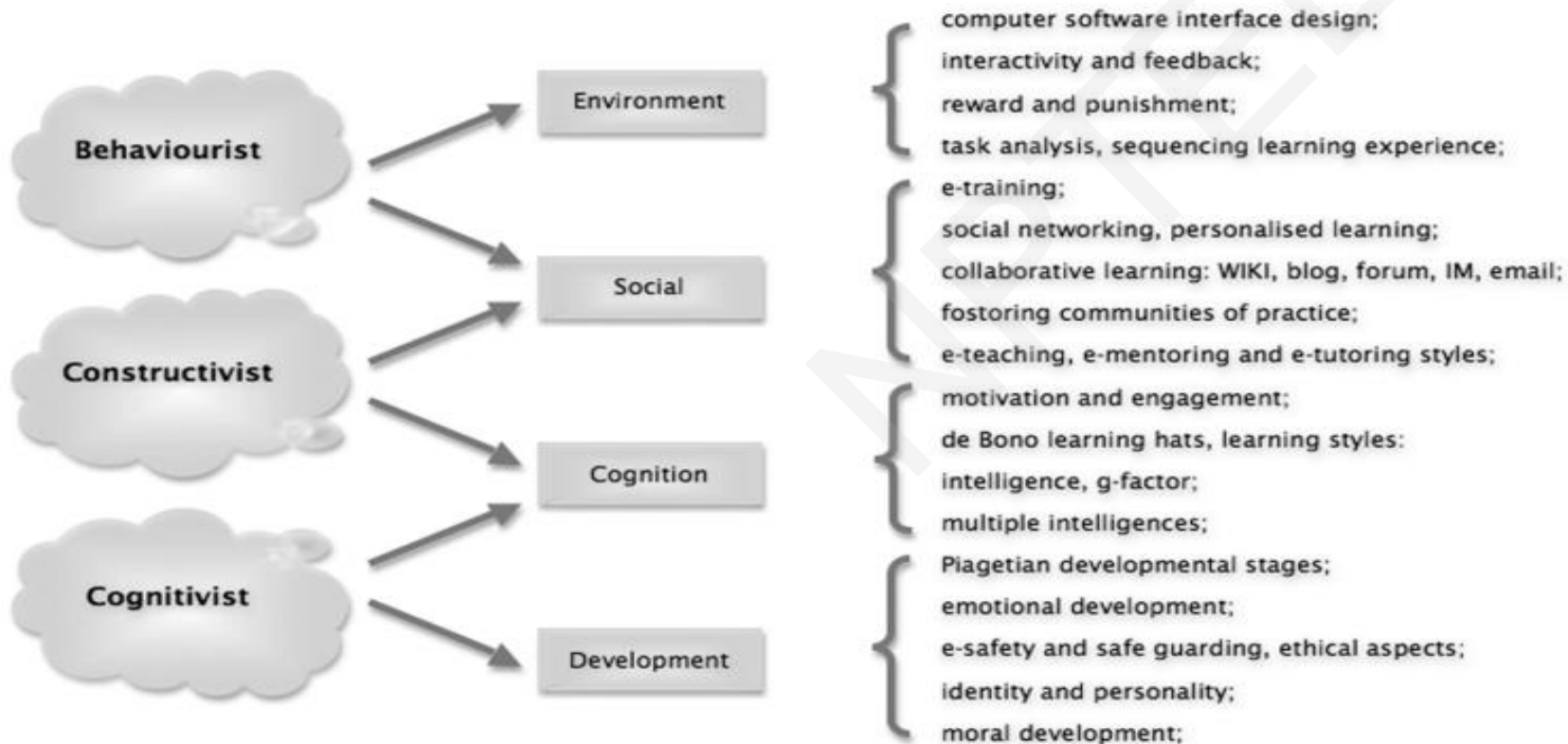
- **Features-**

- **Accessibility, user friendliness** and **social justice**;
- **Personalization, individualization** and learner-centeredness;
- **Open-learning** and **lifelong learning**;
- Bridging the digital divide within society, within families and between generations; and learner expectations
- **Digital Classroom: The changing face of learning**
- A significant number of students expect to do more work at home than in school.
- Many students regard knowledge of 'how to' as more important than 'what'.
- The students' personal computers are the vehicles for learning.

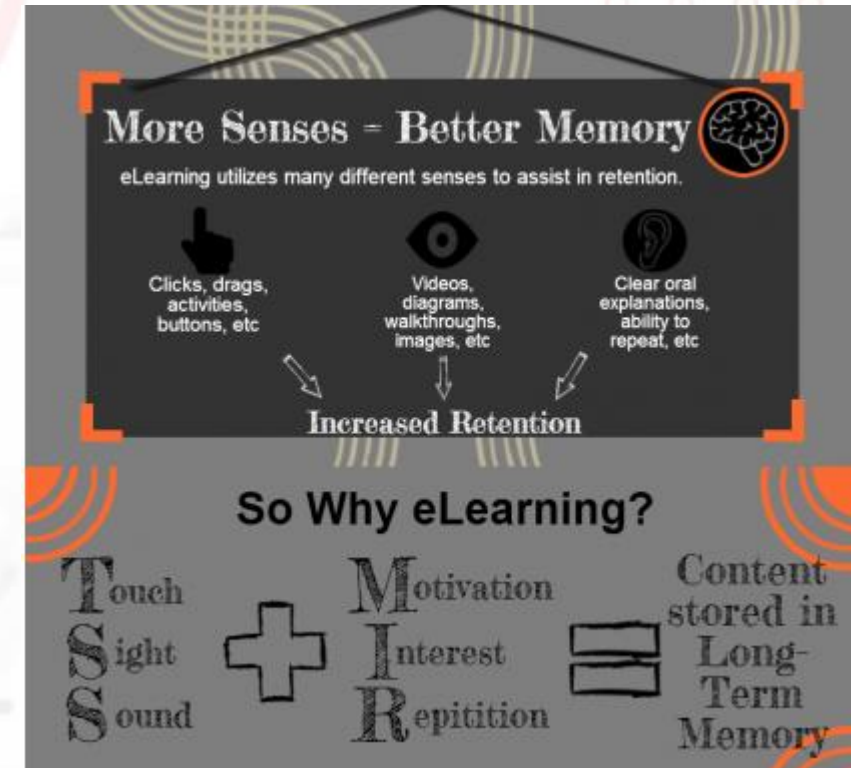
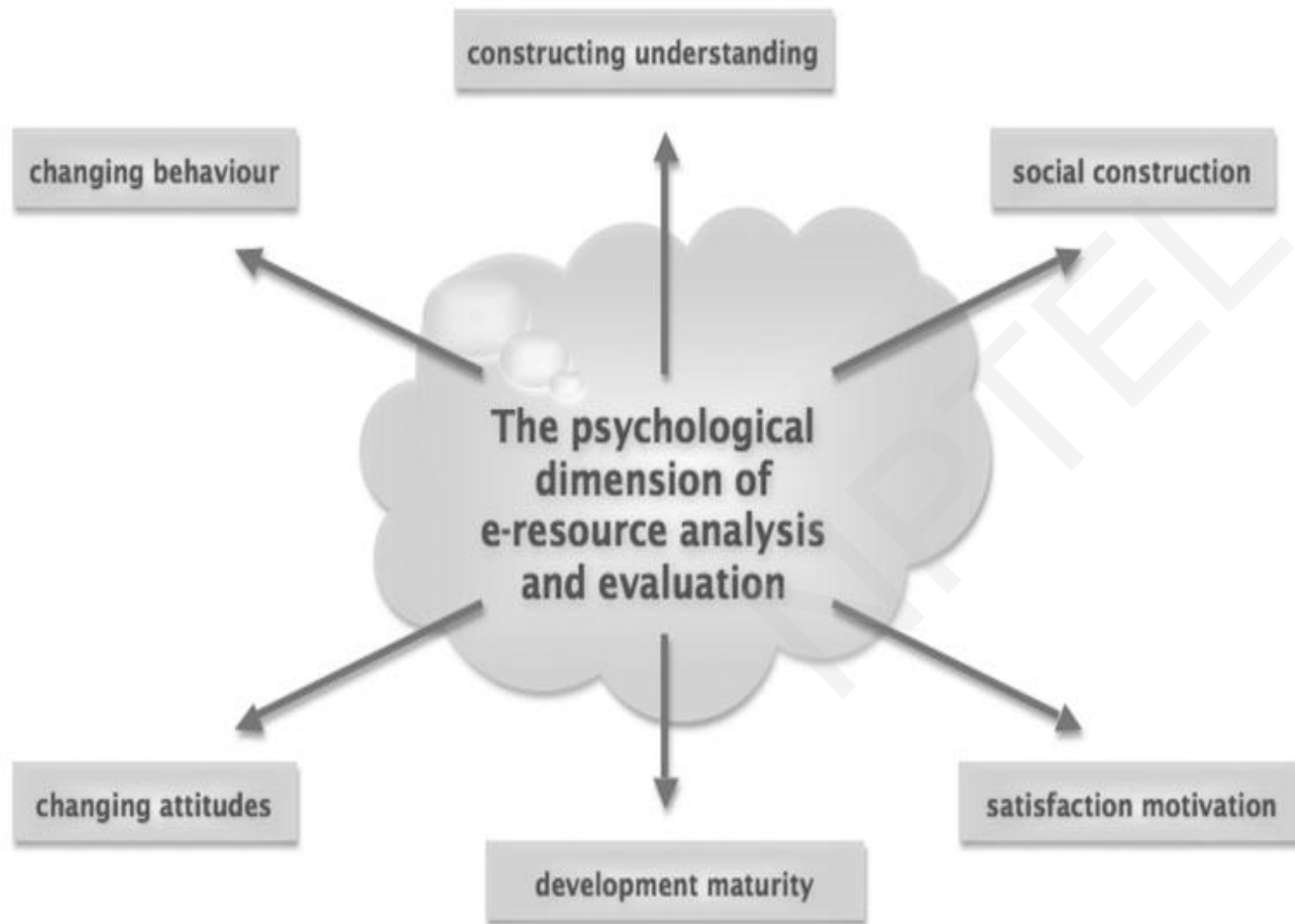


- The students' **patterns of learning** are not found in the classroom.
- The **computer accommodates the range of intelligences** (Gardner, 1983) and preferred **means of learning**, **impacting on students' attainment**.
- Mapping the Psychology of E-learning:

Major learning theories in relation to technology-enabled teaching



- The Psychological dimension of E-resources:





- Characteristics of E-Learning:
- e-Learning is a comprehensive term, which implies computer-enhanced learning.
- It is purely restricted to online learning, undertaken by way of the internet or web-enabled technology.
- It has a wider implication than computer-based learning or computer-aided instruction.
- It is not a synonym of audio-visual learning, multimedia learning, distance learning as these are just complementary.
- It has to be clear that the word 'e-learning' is confined to the form of learning undertaken, facilitated and supported via Web-based instructions and Internet-based communication.
- Types of e-Learning:
- Text Driven: In a text-driven e-learning course, the content is simple and comprises text, images, audio, power-point presentation and text questions.





- **Interactive**: In an interactive e-learning course, interactive components can be found which **enhances the value of learning**.
- It may include graphics, charts, diagrams, videos, etc.
- **Simulation**: It is a highly **interactive** kind of e-learning, which includes graphics, video, audio and games too.
- **Benefits of e-Learning-**
- **Accommodates Everyone's needs**: It **suits everyone's needs**, i.e. be it a working person or housewife or a student, anyone can have access to the information, as per their own pace.
- **Lectures can be taken multiple times**: In contrast with traditional classroom teaching, one can have **access to the content any number of times**, required during revision while preparing for exams.
- **Scalability**: It facilitates in developing and **communicating new training policies, concepts and ideas**, be it for formal education or entertainment.

Benefits of e-Learning

- Accommodates Everyone's needs
- Lectures can be taken multiple times
- Quick delivery of lessons
- Scalability
- Consistency
- Reduced Costs
- Environment-friendly



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Psychology

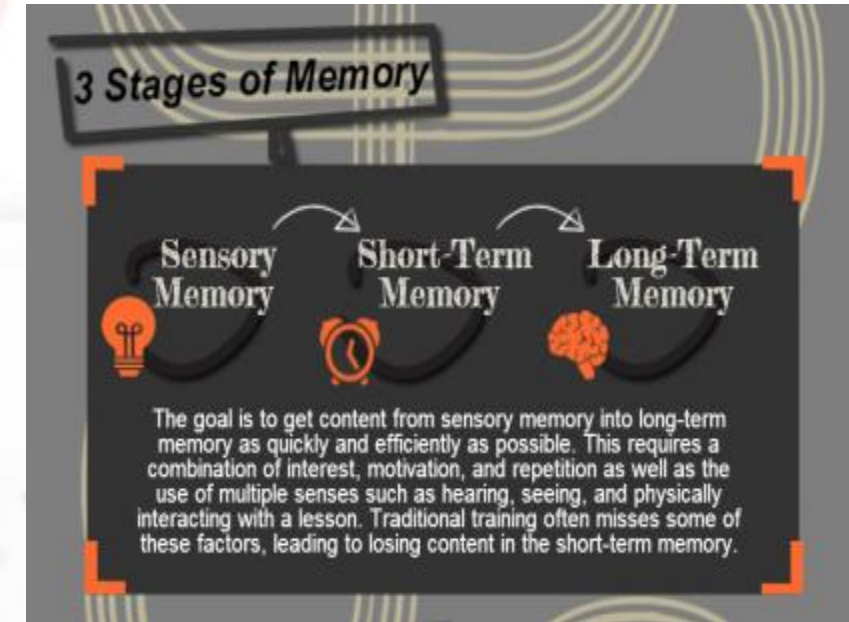


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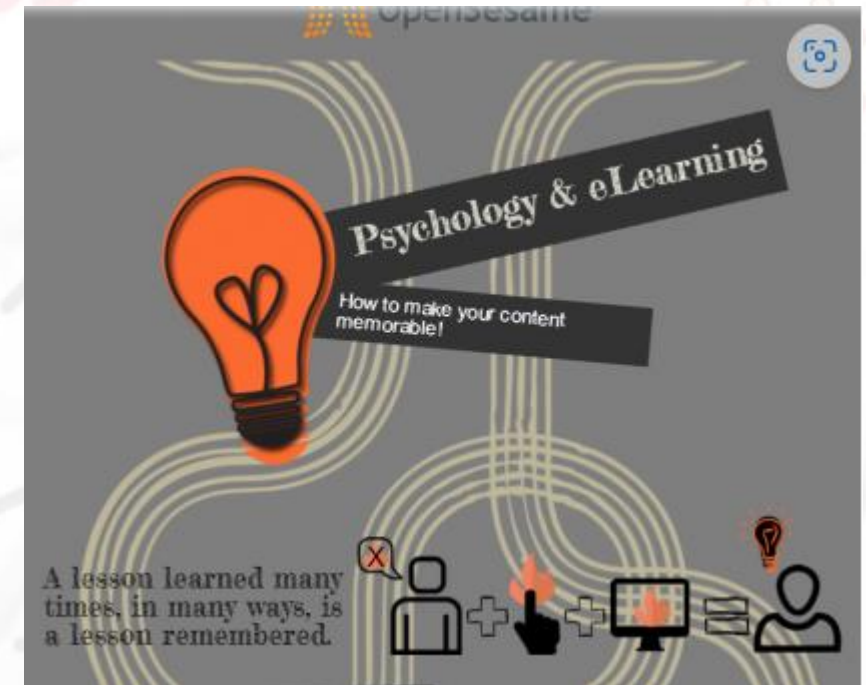




- **Quick delivery of lessons:** It enables quick delivery of lessons as the lessons start quickly and summed up in a single lecture, and the learners can determine their own speed of learning.
- Also, it **saves time** for the students, they don't need to travel to the institution.
- **Consistency:** It tends to enable the teachers to reach millions of students in one go and communicate the message to the target audience.
- Moreover, all the students get the same training at the same time.
- **Reduced Costs:** It is cost-effective, in the sense that people don't need to spend more on travelling, course material, accommodation, and trainers.
- **Environment-friendly:** This mode of learning is a paperless way of learning which protects the environment to a great extent.



- Learners are required to keep themselves motivated.
- It facilitates the learners to incorporate learning around their lifestyles, allowing people to further their career and acquire new skills.
- 'Learners learn more using computer-based instruction than they do with conventional ways of teaching, as measured by higher post-treatment test scores.'
- 'Students who took all or part of their class online performed better, on average, than those taking the same course through traditional face-to-face instruction.'
- 'When faced with learning the same things as learners using other approaches, learners using computer based instruction generally do so in less time.'
- 'The most powerful learner variable is prior knowledge.'
- 'What is emerging most clearly from the technological explosion is, ironically enough, a refocusing on people.'



- Recent thinking in **cognitive psychology** places the individual firmly **at the center** of the learning experience.
- **Individuals create their own** understanding of **knowledge** and the world.
- By interacting with their environment, a **person constructs, investigates** and modifies their **cognitive model** of the world in an iterative fashion.
- **Improving e-learning:**
 - 1. Learning is based on **existing knowledge** and **experience**;
 - The **prior knowledge** and **experiences** that a learner brings with them will play an important part in how they **interpret new knowledge** and make sense of it.
 - It is therefore essential that this prior knowledge is activated when they are learning, by **using analogies, metaphors** and making parallels with this existing knowledge.
 - 2. Learning is **idiosyncratic**





- As learning depends on the learner's prior experience, and **no two learners will have the same experiences**, new information will be dealt with in different ways by different learners.
- This means that **programs must** not only offer different paths through the materials, they should also **offer different perspectives on the subject matter**.
- **Learning is goal-orientated**
- Individuals, and adults in particular, **approach learning** and training with **specific objectives** in mind.
- They need to realize that the **activities** they are undertaking are **related to their real life requirements**.
- And because these needs are often related to solving problems, courses that are task focused situated in environments that help them to activate the knowledge in real life, **problem centered environments** will **result in more effective learning**.





- **Learning depends on self-awareness**
- Good learners know not only **what and why they are learning** but also **how they learn**.
- This lets them take a 'step back' from the actual material and start to make the **relationships with their existing knowledge** and the **application of the new information** to their needs.
- Courses need to build in **self-reflective activities**, through questioning and tasks that make this explicit.
- **Learning is social**
- People do not learn material in an abstract vacuum, they learn it, and use it, in a socially situated context.
- This idea is becoming increasingly important in the most recent thinking about learning, giving rise to the **Social Constructivist School**.
- Incorporating e-mail, on-line discussions, virtual communities and other peer-to-peer communication strategies into the whole learning environment are ways of **facilitating the social dimension to learning**.



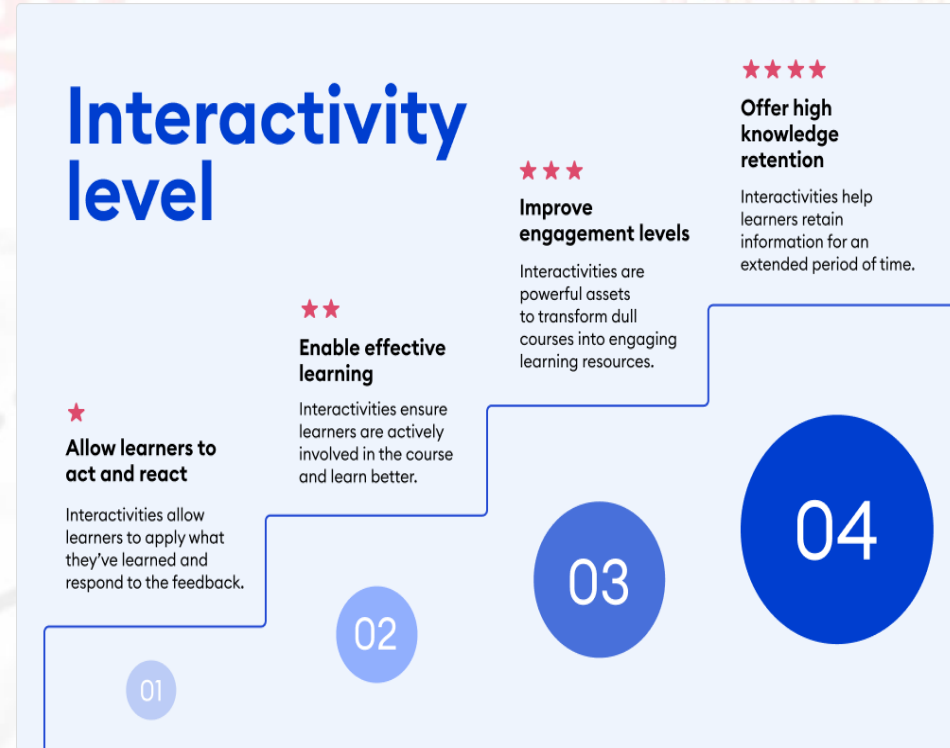


- **Learning is active**
- Developing the **neural networks** that constitute the physical **basis of memory** is an active process.
- Learners that are not **engaging with the learning materials**, in other words actively **cognitively processing** what they are learning, will not have a successful educational experience.
- **Active learning is the crucial determinant of successful E- learning.**
- There are many **dimensions** to **active learning**:
- Individuals' **preferences** for the way that material is presented, their **learning styles**, implementing cognitive strategies for **attention** focusing, **interactivity**, the wider context in which the **learning is situated** and many more.
- However, paramount amongst these elements, and underlying many of them, is the factor of **motivation**.
- In e-learning courses, **multimedia - animations, video, audio** and so on are often cited as promoting interest and motivation, and to an extent this is true.
- But they are still passive media.



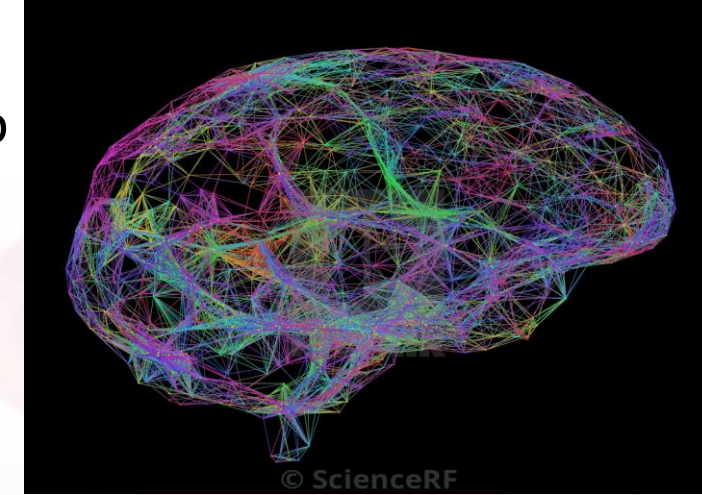


- The key factor is the **meaningful use of interactivity**; interactivity that requires the learner to think.
- Clicking a button is not interactive, nor is dragging an object to another one to start an animation.
- Getting someone **to categorize, analyze, synthesize, problem solve**, and so on, is.
- **Games** can not only **involve problem-solving applications** of knowledge techniques but also elements such as **manual dexterity, humor** supported by artificial intelligence to create such environments.
- The **power of the human mind**-The ability of the brain to process information is phenomenal; The brain can process 36,000 visual cues per hour!
- Research establishes the **capacity of the brain to 'dual process' both visuals and words** and that **capitalizing on** this ability creates **deeper understanding** and greater recall.
- The brain has to work **to expand neural interconnectivity**, and to do that the learner must **actively participate in the experience**.
- A recent study showed that the **brain is more active when it is sleeping** than watching TV.
- Good e-learning has the power to be the equivalent to an hour down the gym for the mind.





- Psychology agrees on the many **benefits** that eLearning offers to learners' well-being.
- The **flexibility** and **convenience** that come with eLearning allow people to expand their horizons regardless of busy schedules or the inability to attend lessons in person due to geographical limitations.
- Additionally, eLearning helps with **knowledge retention** because of the **use of interactive media**.
- **How Does The Brain Learn?**
- **Learning Process-**
- Our brain has approximately 85 billion neurons that send information to one another.
- When we're learning, our **brain makes new connections between its neurons**.
- The more we **practice**, the **more connections** our brain creates.
- So, we can say that when we're good at something, it's usually because our **brain has made stronger connections** with this information.





- More specifically, the **learning process starts with perception**, during which the brain receives information from the environment.
- Then comes **attention**, where the brain keeps only vital information.
- The third step is **memory**, the **brain's ability to store information**.
- There are **different types of memory**, such as **sensory, short-term, and long-term memory**.
- Lastly, there's **retrieval**, during which you can apply everything you know to different situations.
- **Factors That Affect Learning**
 - **Motivation** is one of the primary factors that influence learning because it prompts people to become more and more **involved in the learning process**.
 - **Motivation** can come from **external factors**, like encouragement, rewards, or punishments, and **internal factors**, such as someone's **interest in knowledge**.





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Feedback also affects learning:

- it gives people **insights into how they're performing** so they can make necessary changes in their learning strategy.
- Similarly, **practice helps** people **apply** their **knowledge** and **skills**, whether they do it over a certain period or in a single practice session.
- However, **prior knowledge** is also crucial but not needed.
- It provides learners with a **foundation for learning new things**, allowing them to apply their learning patterns.
- **The Potentially Negative Impact Of eLearning On The Brain-**
- Our **working memory** is temporary and thus **has limited capacity**.
- The **cognitive load** theory suggests that **when we overload** our working memory, **we are unable to acquire** and **process new information**.
- **eLearning may increase the load on our working memory** because we are exposed to multiple media simultaneously and have to switch between different platforms or learning materials.

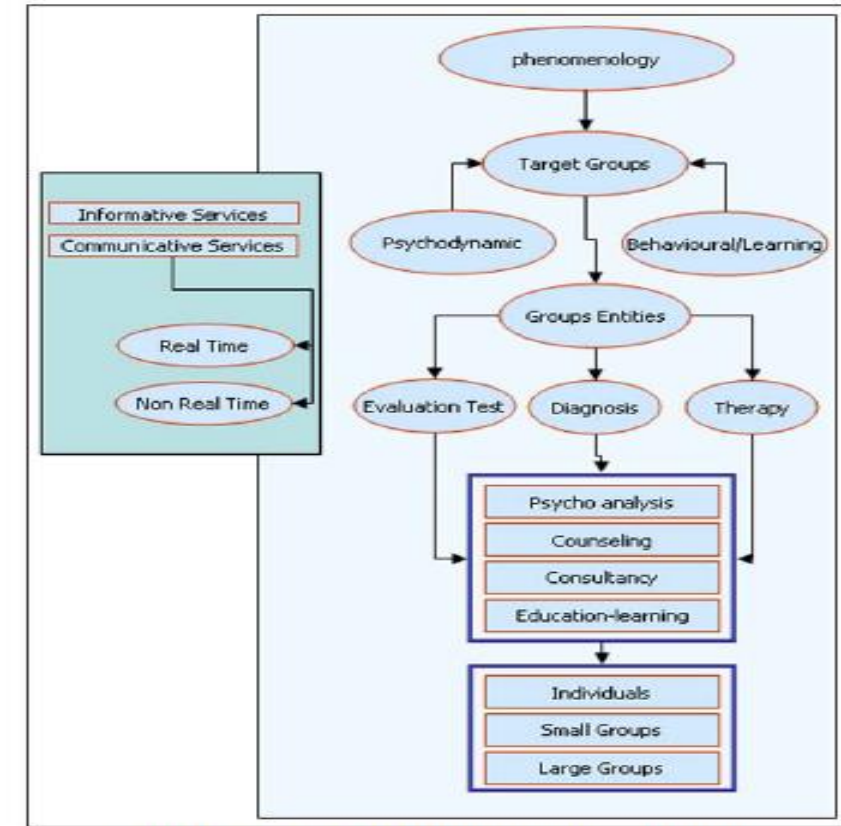


Figure 4. Hybrid psychology structure

- According to a study by the **American Psychological Association**, **multitasking** isn't as convenient as it seems because **shifting between tasks** can cost as much as **40 percent** of someone's productive time.
- **Multitasking** prevents us from focusing on a specific **activity**, resulting in **weaker connections** between our **neurons**.
- Learners may multitask when engaging in an online course, resulting in **shorter attention spans**.
- **Navigating The Challenges**
- **Reducing Cognitive Load-**
- The way we learn is highly influenced by our **mental effort**.
- So, **online courses** should be designed to minimize it, allowing us **to learn more effectively**.

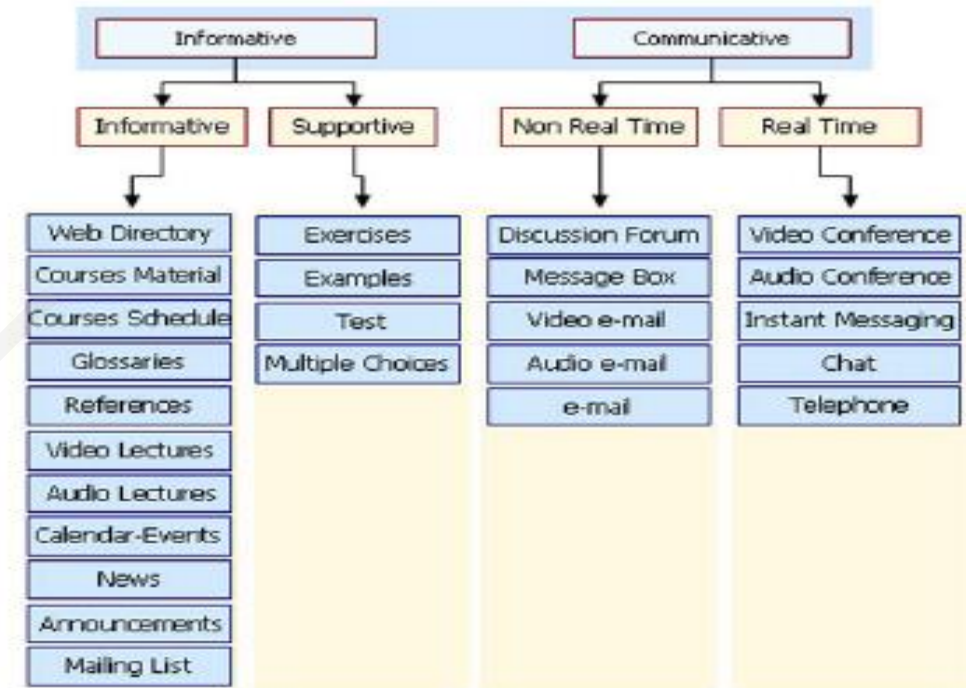


Figure 6. Informative and communicative tools



- To start, **use plain language** when **designing eLearning material**, and avoid presenting long pieces of text.
- Instead, **use bulleted lists, headings, and categories**.
- Additionally, when using multimedia content, **select relevant videos, infographics, animations, etc.**, and **strategically implement** them not to distract learners.
- Most importantly, be alert and **offer guidance and constructive feedback** when needed.
- **Encouraging Active Learning**
- Learners who **actively participate in online lessons** are more likely to retain knowledge and achieve their goals.
- To help them with that, give them constant **opportunities for problem-solving** through realistic scenarios or **case studies**.
- We can also ask them to hand in **essays in which they will reflect on** what they've learned and help them remember it.
- **Discussions** and **collaborative work** can also encourage a proactive role, especially when **paired with games** and other **interactive activities**.





- **Social Interaction**-when learners interact with each other online, their messages can be misinterpreted and lead to miscommunication.
- **social media, group chats**, and **video conferencing** can be used to bring everyone together from afar.
- **Encoding**-The **brain** often **benefits from physically writing in a notebook** since it contributes to encoding.
- Hence, learners can also try **keeping notes while attending online courses** to speed up the encoding process.
- We can even encourage them to **keep an online journal to track their progress** and **outline the key takeaways**.
- **Immediate feedback**- Instructors in online courses, though, can make full assumptions only after they've received the **data analysis of learner performance** and can make posterior **suggestions** and **comments**.
- That said, there are ways to provide **immediate feedback via simulations, scenarios**, and other real-world activities.

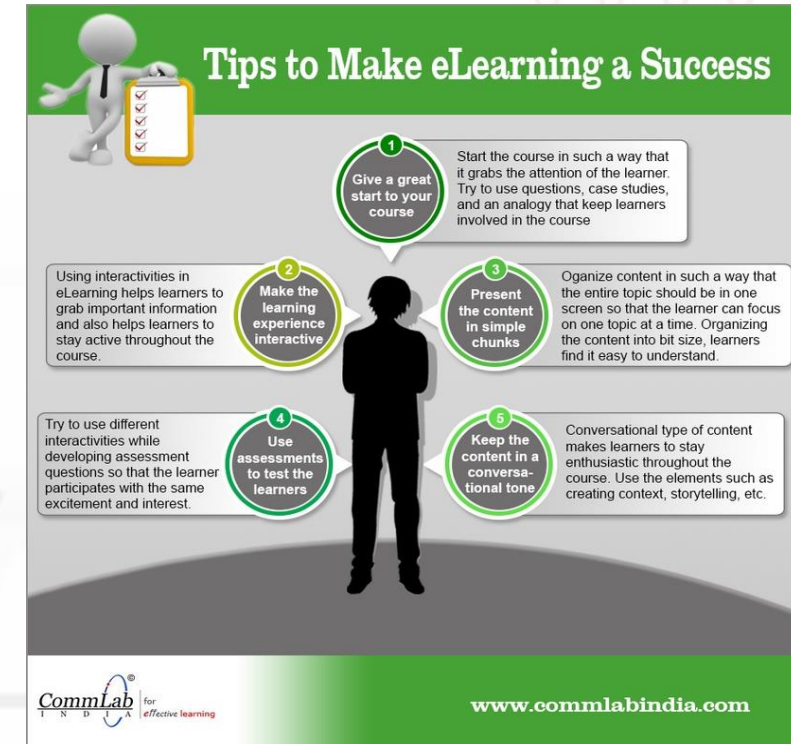


- **How to make e-learning Effective ?**
- **Adopt blended mode** — combine the Internet education with the traditional one.
- Put **more stress on interaction** with the students than on preparation of sophisticated automatic tools.
- **Make a social group** out of e-course participants in co-operation with an integrator and assistants.
- **Personalize the course** requirements by adjusting the level of its difficulty to the participants.
- **Set-up clearly defined goals** and do not overload the syllabus.
- **Use multimedia** as widely as possible and, in particular, record the sound files.
- **Plan the student's activity**, check the level of knowledge acquisition but try to grade the tests automatically.
- **Separate the teaching process** from **the knowledge certification** — let the students repeat the tests as much as needed during the learning stage.
- **Monitor the educational process** — in the evaluation questionnaires ask open questions; **Be flexible**.





- Education should not only be a transaction of knowledge between educators and students, but also has an objective to enhance students to become qualified human.
- In the psychology perspective, to realize it, the educational curriculum should consider every individual is unique in the intellectual, attitude, motivation, and feelings.
- eLearning is good system of education which gives freedom to the students to learn, based on their uniqueness.
- However, some critical aspects must be investigated to develop a successful eLearning by paying attention on the characteristics of each individual.
- There are five critical aspects of psychology which are needed to be taken for guarantee the implementation of successful eLearning.
- There are motivation, emotion, self-discipline, social skill, and stress. By involving those aspects, the eLearning would not only be one way of transferring knowledge from teachers to students.
- It will be more interactive, more interesting, and would help the eLearning participants to follow the learning process.



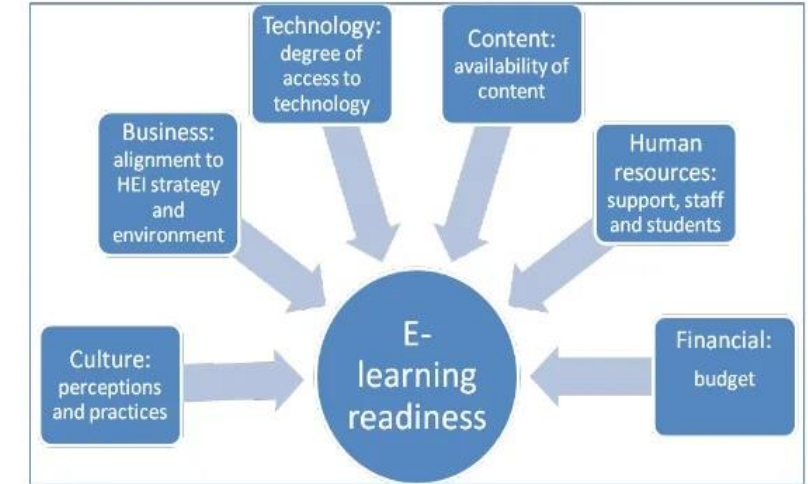


- Teachers' E-Learning Readiness:
- **Psychological Readiness**-bears an importance to determine and **assess** their **mindset** and **attitude**.
- it is important **to change the mindset** of everyone involved in the teaching and learning process.
- **Technical Skill Readiness**- is an important factor that **influences the success** implementation of e-learning.
- **Demographic, Socio-economic & motivational factors** of the teachers are also important **factors** for E-learning instruction.
- In **Virtual learning** environment the **usability criteria** refers to-
- **Learnability** refers to the speed and ease with which a novice user can achieve proficiency with the system.
- **Efficiency** refers to the degree to which the system supports the performance of an experienced user in the shortest amount of time and with the fewest steps.



- **Memorability** refers to the degree to which a user, particularly an intermittent or casual user, **can remember how to accomplish a task using the system**, the steps of which were learned previously.
- **Errors** refer to the number of mistakes and **missteps** made by users.
- **Satisfaction** refers to the users' overall **emotional experience** when using the system.
- The **use of topic maps** to create a **user-centered, navigable index of instructor-annotated essays** needs to not only be grounded in theories of educational psychology, but also, necessarily, those of **information retrieval**.
- **Topic map is a learning tool**, because the **explicit navigation** of the topic and association types might **enable meaningful learning**.
- Topic Maps also **reduce the cognitive loads** caused by E-learning content and constant attention.

Theoretical model





- Boosting Learner Engagement With E-Learning Content-
- 1. Prior Knowledge Affects Learning-
- Start with an assessment of baseline level; use those results to identify strengths and weaknesses and formulate the training material in a way that is engaging and non-repetitive for learners.
- 2. Learning By Doing-
- Learners are able to retain more information when they are active participants in the learning process.
- This would mean engaging in hands-on activities, simulations, group discussions and quizzes, and any other activity that gets them moving and reflecting on the subject at hand.



• 3. Driving Motivation-

- One is **explaining** the importance and **real-life application** of the training ;
- Another useful tool is **motivation** through **goal-setting**; can encourage learners to **set personal mastery goals** so that they feel like they're working towards an achievement that is important to them.

• 4. Keeping It Social-

- Create a **training environment** that **promotes collaboration** and **teamwork**;
- observe learners' benefit from the **insights** and **diverse opinions** of the team;
- Just remember to encourage **healthy competition**, cultivate a **culture of collaboration**.





- **5. Providing Feedback-**

- Feedback allows learners to **assess** their **performance** by identifying their strong suits and **areas for improvement**;
- Instructor has to offer **clear, positive**, and **time-appropriate advice**;
- Thus, helping them to apply it and **make a positive change** in their learning progress.

- **6. Promoting Self-Regulation-**

- **Self-regulation skills** such as organization, **self-control**, **attention**, **memory strategies**, and encouraging learners to **set personal goals** can be encouraged through **modeling / direct instruction**.
- **7. Develop learning goals, performance goals**, and learning objectives in your eLearning course-
- Be as **specific** and **detailed** as possible to **let the learners know exactly which skill sets they'll be developing**,





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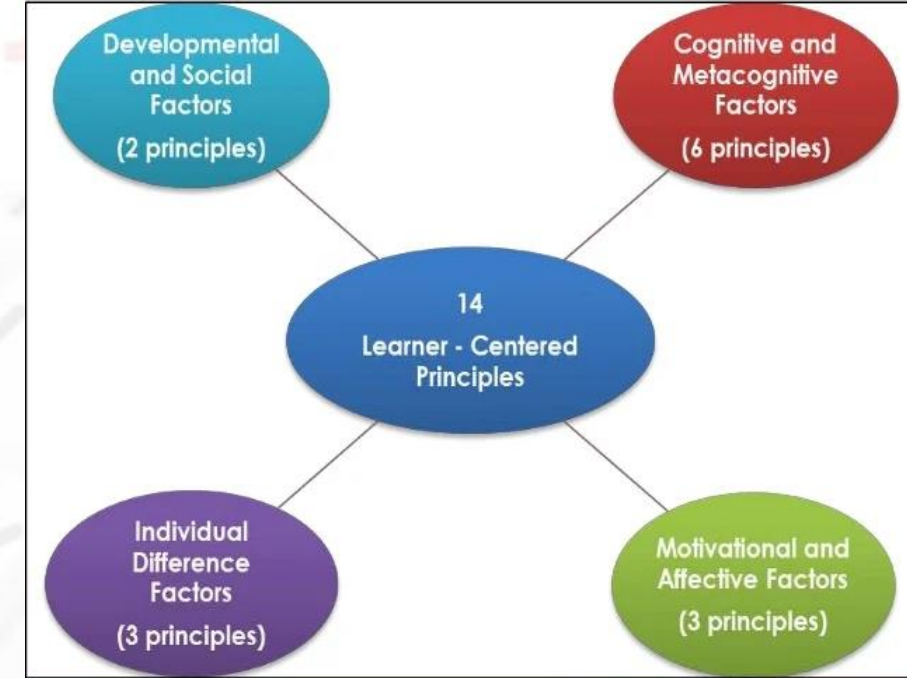


- what key pieces of knowledge they will be absorbing?;
- what they can do with all of this newly acquired information?
- 8. People have an inherent need to learn and experience new things-
- Can tap into the power of curiosity and self sufficiency by creating activities, like eLearning scenarios and eLearning games, that encourage learners to think for themselves.
- Another key component of this particular psychological principle is experimentation.
- Designing activities that require learners to apply already learned knowledge to solve problems allow them to increase their comprehension of a given topic.
- This may come in the form of group exercises wherein learners must create blogs that discuss a particular topic, or develop an eLearning presentation that delves into an idea or theory that they must present to their peers.





- Applying Psychology Principles before Developing an e-Learning Course-
- Taping the power of the subconscious for eLearning courses that would immerse, engage, and inspire the learners;
- If learners can't see the real world applications, they won't see the value of the eLearning course;
- Hence we need to develop learning goals, performance goals, and learning objectives in our eLearning course;
- To be as specific and detailed as possible;
- Let learners know exactly which skill sets they'll be developing, what key pieces of knowledge they will be absorbing, and what they can do with all of this newly acquired information.



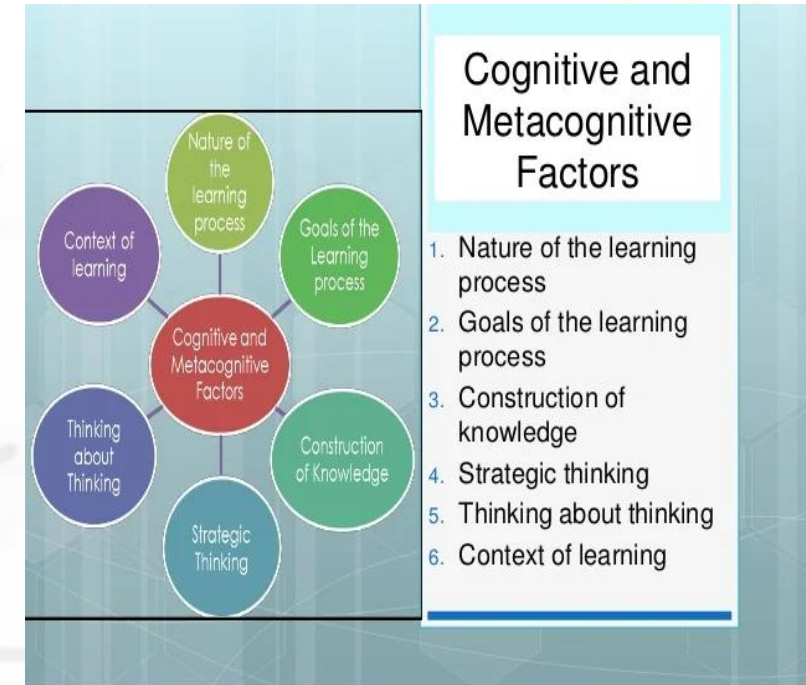
- Positive reinforcement is a powerful tool.
- This positive reinforcement may be as simple as praise or feedback after they've completed a task successfully.
- We can also design a game in which learners level-up after they've scored high enough on a module.
- All of these things give them the immediate rewards that they need (the instant gratification), as well as the long term benefits they'll receive by acquiring new knowledge or skills.
- Learners gravitate toward collaborative experiences.
- There are a variety of ways to integrate social interaction into eLearning environments.
- Technology allows us to create collaborative experiences that enable learners to benefit from the experience and skills of their peers, even if they are a world away from each other.

Dispositions to Learn

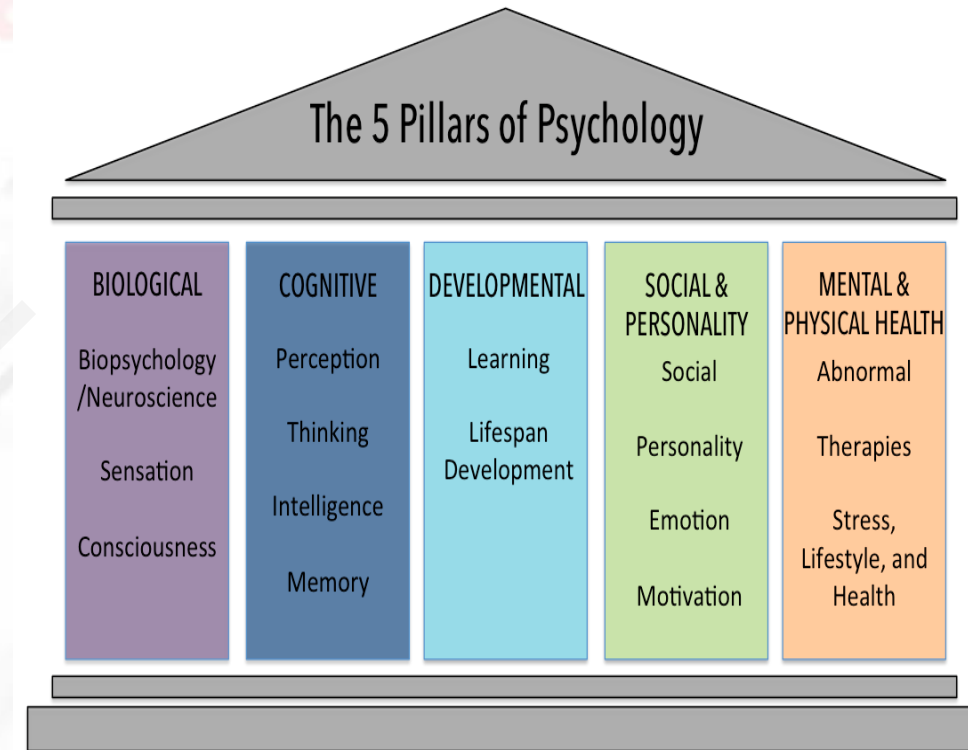
- independence
- creativity
- Self - motivation
- resilience



- We can design **collaborative experiences** by using social media sites in your eLearning strategy, such as **Facebook** and **LinkedIn**.
- Learners can form groups with other **like-minded peers** or interview **professionals** to learn more about a particular subject.
- Learners need to be able to **relate to the information**.
- One of the most useful ways to integrate this psychological principle into your eLearning course is **to make it relatable**.
- To ensure that you've **thoroughly** researched and **analyzed** your **audience**.
- This is the only way that we'll be able to tap into their emotions and build a connection with them.
- Learn all that you can about their **cultural, professional, and educational background**, so that we can make the **content** relatable and **relevant for them**.



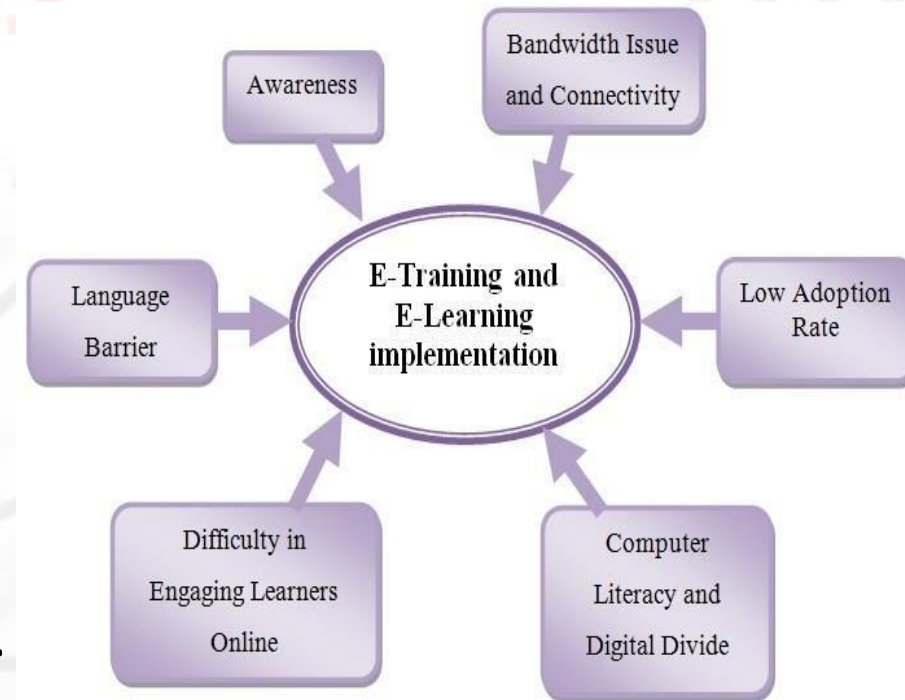
- Learning is, ultimately, governed by **psychology principles**.
- Our **brain** and our **learning behaviors** dictate how and why we learn.
- **Storytelling** is an extremely useful tool for learning and memory and is one of the many different techniques used in e-learning.
- Stories, pictures, experiences; these are the things we remember.
- Remembering requires interest, motivation, and repetition; **we have to see it, hear it, touch it, enjoy it.**
- In the end, storing something in memory takes work and effort, so we have to make your information worthy of remembering.
- **E-Learning is revolutionary**, giving us the opportunity to design more effective training.





- Factors for ensuring optimum implementation of e-Learning-
- **Motivation** -has influences for a person and drive he/she can success or not in doing something.
- Motivation also has a control **to push the individuals for starting** or continuing their efforts .
- eLearning also requires the students to have a strong motivation if they want to succeed in the learning process.
- **Strong self-motivation** is needed for eLearning students .
- **Emotions**- have an important role in the process of learning.
- **motivation** together with the **emotions** can increase sharply and eventually produce a **good performance**.
- When the students succeed in eLearning, they will display high **enthusiasm** and **pride**.

So, the learning system must able to create positive emotions.





- **Self-Discipline-** eLearning system gives the freedom for students to study anytime and anywhere.
- Individuals are the main actors here; It needs self-discipline from the student to follow and manage their study.
- If someone succeeds with eLearning system, it can be assumed he/she has an internal discipline.
- **Social Skills-** Implementation of eLearning has to pay attention on social skills and behaviors (good habits, ethics, and psychology).
- The skills aspect of empathy, sympathy, and skill of listening skills can be learned from the personal relationships and teachers - students interactions.
- eLearning has to provide the collaboration system between participants and discussion between student and the teacher.

The Ten Critical Success Factors for an eLearning Strategy



- **Stress**- A person will **experience stress** in the learning process, especially **if they learn only in one way** or individually.
- In a traditional learning system, there is the possibility for a teacher to interact with the students **face-to-face to increase their motivation**, become a place of counseling, and discuss their idea. (Source: International Conference on eLearning for Knowledge-Based Society, 16-17 December 2010, Thailand).

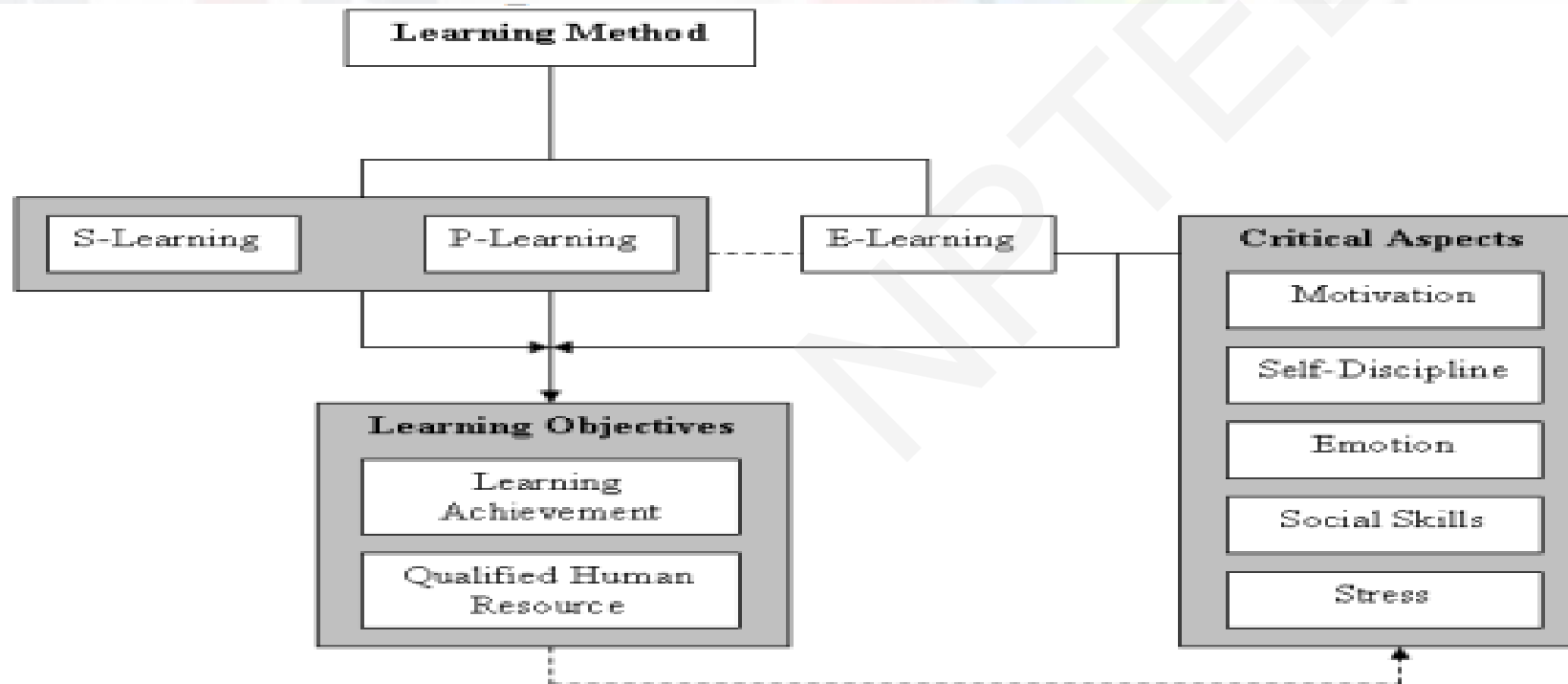
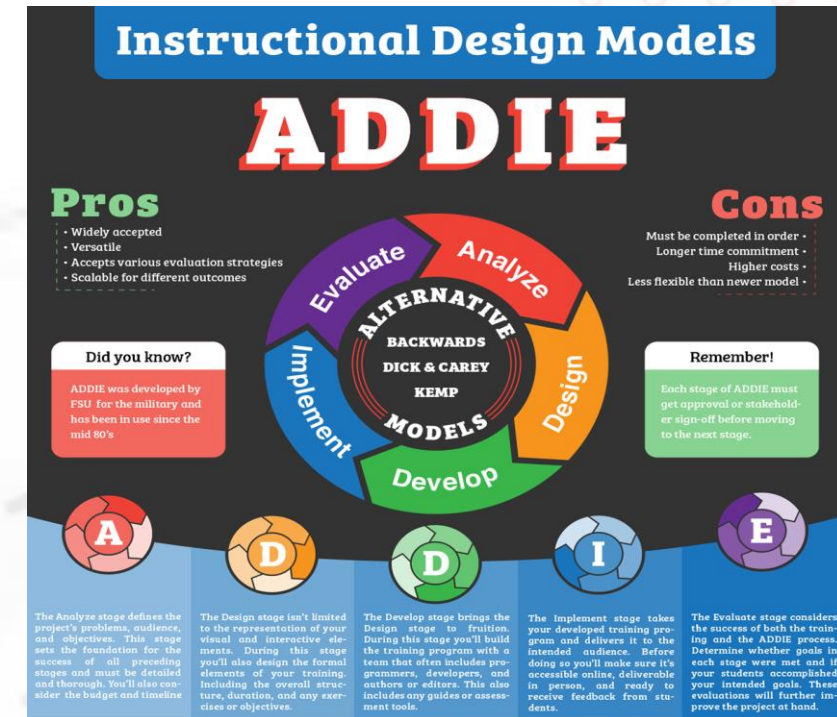


Fig. 1 The influence of the five critical aspects on Learning



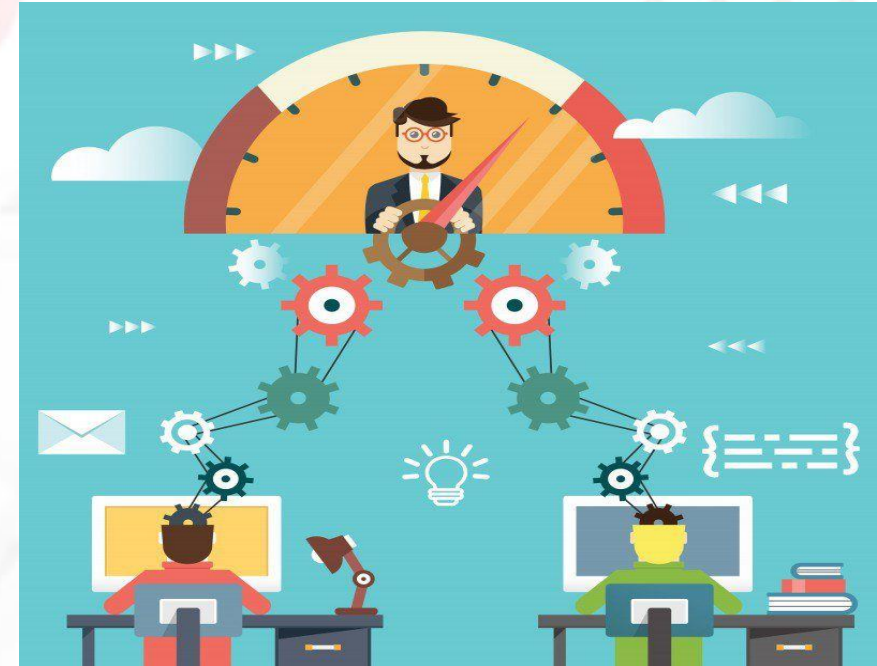
• Instructional Design For eLearning-

- Today, programmed instruction can be perceived as an **adaptive hypermedia systems applied to online education** with behavioristic elements, the resemblance of which with programmed instruction is very difficult to be ignored;
- We tend to "cover" them in **modern "constructivist"** packages claiming that we give learners "**freedom of choice**."
- Only freedom **learners are allowed to choose** & follow the predefined learning paths, to **carefully track** and **follow programmed Instructional Design** of really good eLearning courses.
- **Instructional Design** for eLearning based on the **behavioristic approach** starts from the basic assumption that knowledge is objective, meaning that there is **only one correct answer** to give or a specific approach to follow.
- **Techniques To Be Used For The Instructional Design-**



Psychology

- **Discrimination**
 - When the learning objective requires learners to identify whether a concept belongs to a specific category or not.
 - Learners should be able to identify key characteristics and qualities of the category and judge whether the new information shares the same qualities so as to belong to the group or not.
 - Drag-and-drop exercises to classify concepts into different categories may serve as examples of developing discrimination activities in terms of eLearning course design.
- **Generalization**
 - It's suitable whenever the learning objectives are such that learners, after identifying the attributes of an item belonging to one category, are expected to assign the same attributes to all items within the category.



- Teaching through examples is based on an **inductive approach** of **presenting eLearning content**;
- learners after observing a series of independent online examples should be able to identify their **common characteristic/s** and **generalize by formulating the rule**.
- generalization is very **close to** what today we perceive as **active learning**, with an obvious **direct connection** to **constructivism**.
- **Association**
- **Association** is an example of **conditioning**—which is linking a specific stimulus to a specific response—there is a tendency in today's eLearning **to create eLearning interactions based on association**, whenever the **learning objectives require the new information** presented to be **linked to specific practical applications** of it.
- Presenting **information within context** can help learners **create associations**.





• Chaining

- Chaining is referring to learners' automatic performance on specific procedures with predetermined steps to be followed.
- Like a chain, one step leads learners to the next.
- Performing drill tasks is an example of eLearning objectives that can be mastered through chaining.
- Creating drill tasks in eLearning involves presenting the theoretical model first and then asking learners to repeat the procedure by actually repeating the steps involved in the same order these were presented.
- Through repetition and online practice, learners at the end are able to reach the desired outcome by following the steps exactly as presented in the model.
- Sequence ordering exercises through drag-and-drop are very frequently used in the quiz-making templates most eLearning authoring tools provide.
- Such exercises may serve as typical examples of drill tasks applied in eLearning course design.

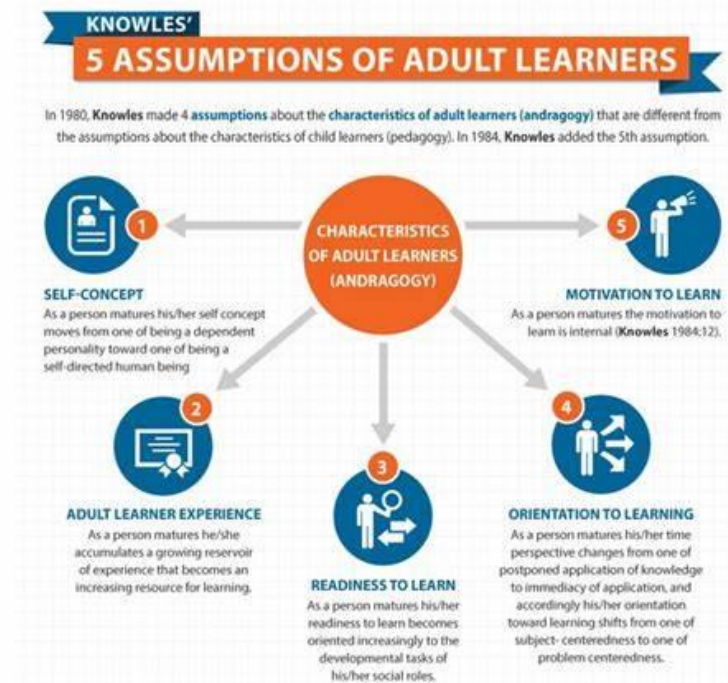


- Reinforcing Desired And Weakening Undesired Behavior-
- What is the instructor's role in this process then?
- Instructional Design for eLearning based on a behavioristic approach sets the type of reactions to be received by learners after interacting with the online training material.
- These reactions come from an online instructor or corporate trainer in synchronous eLearning settings, or from the eLearning course itself in the case of asynchronous eLearning.
- Depending on the type of the reaction received, the learners' expressed behavior may be reinforced or weakened.
- This highlights the importance of feedback throughout the entire learning process, not only at the end of the eLearning course but each time a learner interacts with the system.





- Application Of Behavioristic Principles In Gamification And Game-Based eLearning-
- Gamification is presenting the learning material by employing game design elements in an entertaining way in order to motivate and engage the audience throughout the learning process.
- In gamification, as well as in other types of eLearning activities, the reinforcement of a certain desired behavior can occur in two ways:
- Either by eliciting from learners' particular learning outcomes and rewarding these outcomes by assigning points, grades, badges, a higher position in leaderboards, etc., or by removing from learners specific benefits.
- Applying Adult Learning Theory to eLearning-
- According to the renowned American educator, Malcolm Knowles there are 5 assumptions concerning the characteristics of adult learners, and 4 principles concerning adult learning (andragogy).
- Applying Knowles' 5 Adult Learning Theory Assumptions to eLearning-

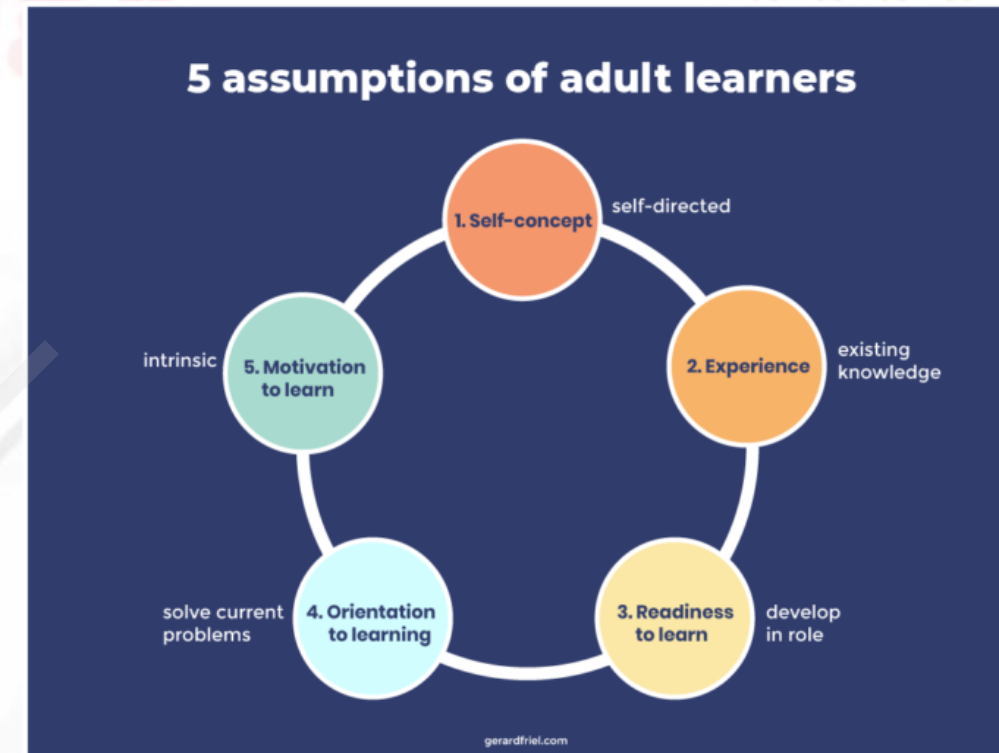




Assumption #1 (Self-Concept)

- Create learning experiences that offer minimum instruction and maximum autonomy.
- A major aspect of designing adult eLearning courses is having an eLearning support system to offer guidance and help, while still giving the eLearning tools and resources they need to learn on their own terms.
- Adult learners acquire new information and build upon existing knowledge much more effectively if they are encouraged to explore a topic on their own.
- Assumption #2 (Adult Learner Experience)
- Include a wide range of instructional design models and theories to appeal to varied experience levels and backgrounds.
- Adult learners are more mature.

- Therefore, they have had more time **to cultivate life experience** and typically have a **wider knowledge base**.
- adult learning audience is going to be more diverse, especially in terms of backgrounds, experience levels, and skill sets.
- While one adult learner may be well versed on how to search for resources online, another may have very little experience using the Internet.
- It's often **best to include a variety of different instructional design models** and theories into your eLearning course or module.
- **Assumption #3 (Readiness to Learn)**
- Utilize social media and **online collaboration** tools to **tie learning to social development**.





- As we get older, we tend to gravitate more toward learning experiences that offer some sort of social development benefit. (For example, we are often more ready to challenge ourselves with new learning opportunities if we know it will help us to fine tune skills that pertain to our social roles.).
- From an eLearning professional point of view, **social media** and **online collaboration tools can help** you to incorporate this assumption into our deliverables.
- **Assumption #4 (Orientation to Learning)**
- Emphasize **how the subject matter is going to solve problems** that an adult learner regularly encounters.
- Adult learners, essentially, need to know the why and when before they actively engage in the eLearning process. (For example, they will not only want to know why they need to acquire specific information, but whether or not that information can be applied in the immediate future. Younger learners accept the fact that the knowledge they're acquiring today may not be used for quite some time.)





- Mature learners prefer to engage in eLearning experiences that help them to solve problems they encounter on a regular basis (in the here-and-now, rather than the future).
- **Assumption #5 (Motivation to Learn)**
- There must be a **valid reason behind** every **eLearning course**, module or educational activity.
- Motivation is key with adult learners; by offering them a **reason for every eLearning activity**, assessment, or eLearning module they'll need to complete.
- eLearning professionals must explain why a particular eLearning course is being taught and why an adult learner must participate in an eLearning activity, in order for the **overall eLearning experience to be meaningful and engaging**.
- While younger learners won't need to necessarily know the reason why they are required to participate in an activity, **adult learners need to feel** as though they are **more involved in the process of learning**.

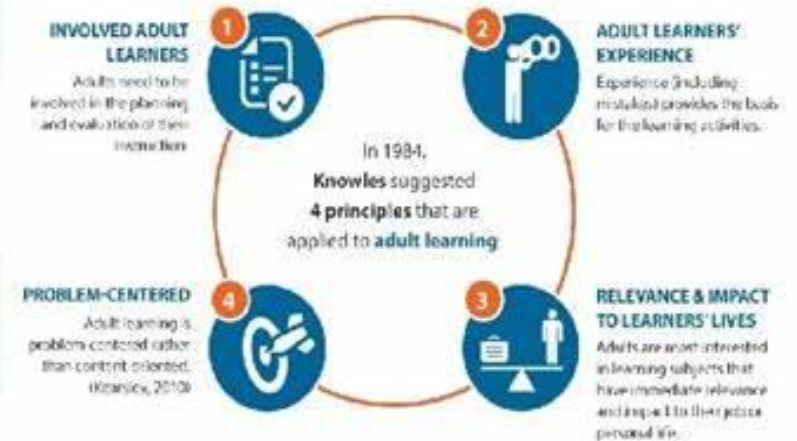
Types of Assumptions

Causal – if I use this teaching approach it will engage students / learning / illustrate complex reasoning (e.g. Modeling critical thinking helps students learn to think critically)

Prescriptive – what a good teacher or class should involve (e.g. a good teacher uses a mixture of teaching approaches)

Paradigmatic – structuring, framing assumptions we take for granted regarding the nature of teaching (e.g. teaching is a trajectory of improvement)

Knowles' Adult Learning Theory



Source: <http://elearninginfographics.com/adult-learning-theory-andragogy-infographic/>

- Applying Knowles' 4 Adult Learning Theory Principles to eLearning-
- Principle of Andragogy#1
- Adults **must have a hand in the design** and development of their learning experience.
- Both adult and younger online learners must feel that they are playing an active role in their own eLearning experience;
- They must truly be an integral part of the development and implementation of the curriculum, as well as of the evaluation process.
- Principle of Andragogy#2
- Experience should be at the root of all eLearning tasks and activities.



- In the eLearning experience that is gathered through instruction and activities, **offer memorization tasks**, create projects and exercises to encourage adult learners to explore the subject matter, and gaining experience.
- By doing this, **adult learners can learn from their errors** and **master their skills sets through first-hand experience**.
- **Principle of Andragogy #3**
- **Real life applications** and **benefits must be tied** to the eLearning course.
- Adult learners need to be able to tie the subject matter to **real world benefits** and **applications**.
- They want to explore, how a particular eLearning course is going to apply to real world situations;
- eLearning professionals can **increase engagement by integrating real life scenarios/case studies** into adult eLearning courses.

Strengths of Knowles Theory

- A set of **core** adult learning principles
- Can be applied to **all** adult learning situations



Psychology

Adults' Theory of Adult Learning

- Principle of Andragogy #4
- Give adult learners the **opportunity to absorb information**, rather than memorizing it.
- The e-learning **content should be problem-centered**, as adult learners' would want to immediately see the impact of the instructions in solving the **real life issues**.
- Thus, the subject matter should offer them the chance **to fine tune skill sets** and **acquire** (and retain) **practical knowledge by doing**, rather than just memorizing.
- Create **activities** that allow adult learners **to delve into specific tasks**, such as **simulations**, that enable them to store the information in their long term memory **through repetition** and **experience**.

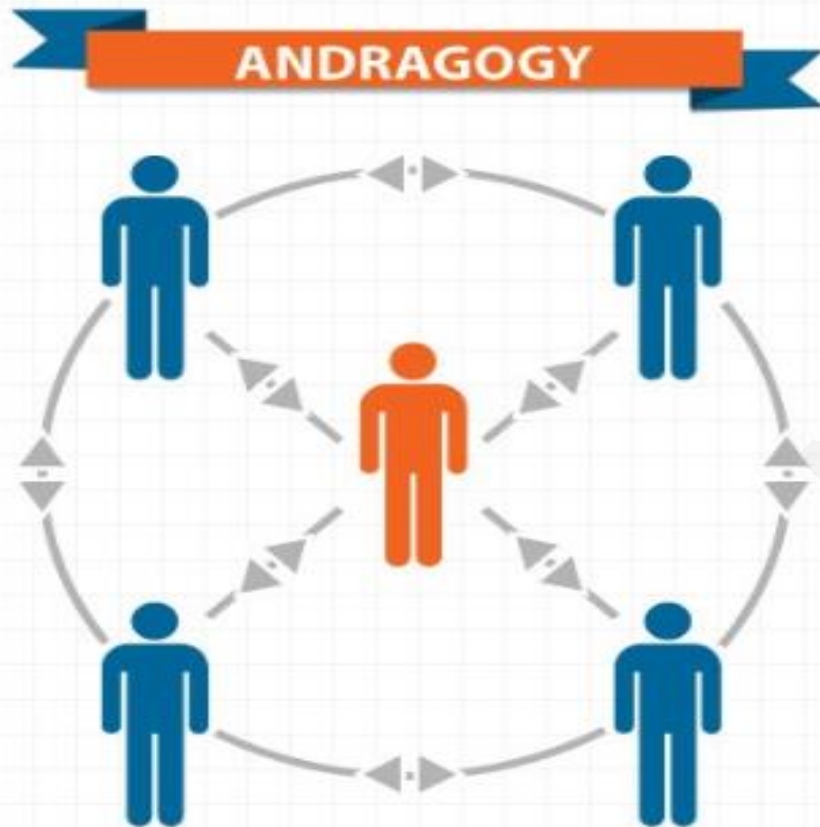




Malcolm Shepherd Knowles (1913 – 1997)

Was an American educator well known for the use of the term **Andragogy** as synonymous to the **adult education**.

According to Malcolm Knowles, **andragogy** is the art and science of adult learning, thus **andragogy** refers to any form of **adult learning**. (Kearsley, 2010).



The term **andragogy** can be supposedly equivalent to the term pedagogy. **Andragogy** in Greek means man-leading in comparison to pedagogy, which in Greek means child-leading. However, it should be noticed that the term pedagogy is used since the Ancient Greek times while Alexander Kapp, a German educator, originally used the term **andragogy** in 1833.



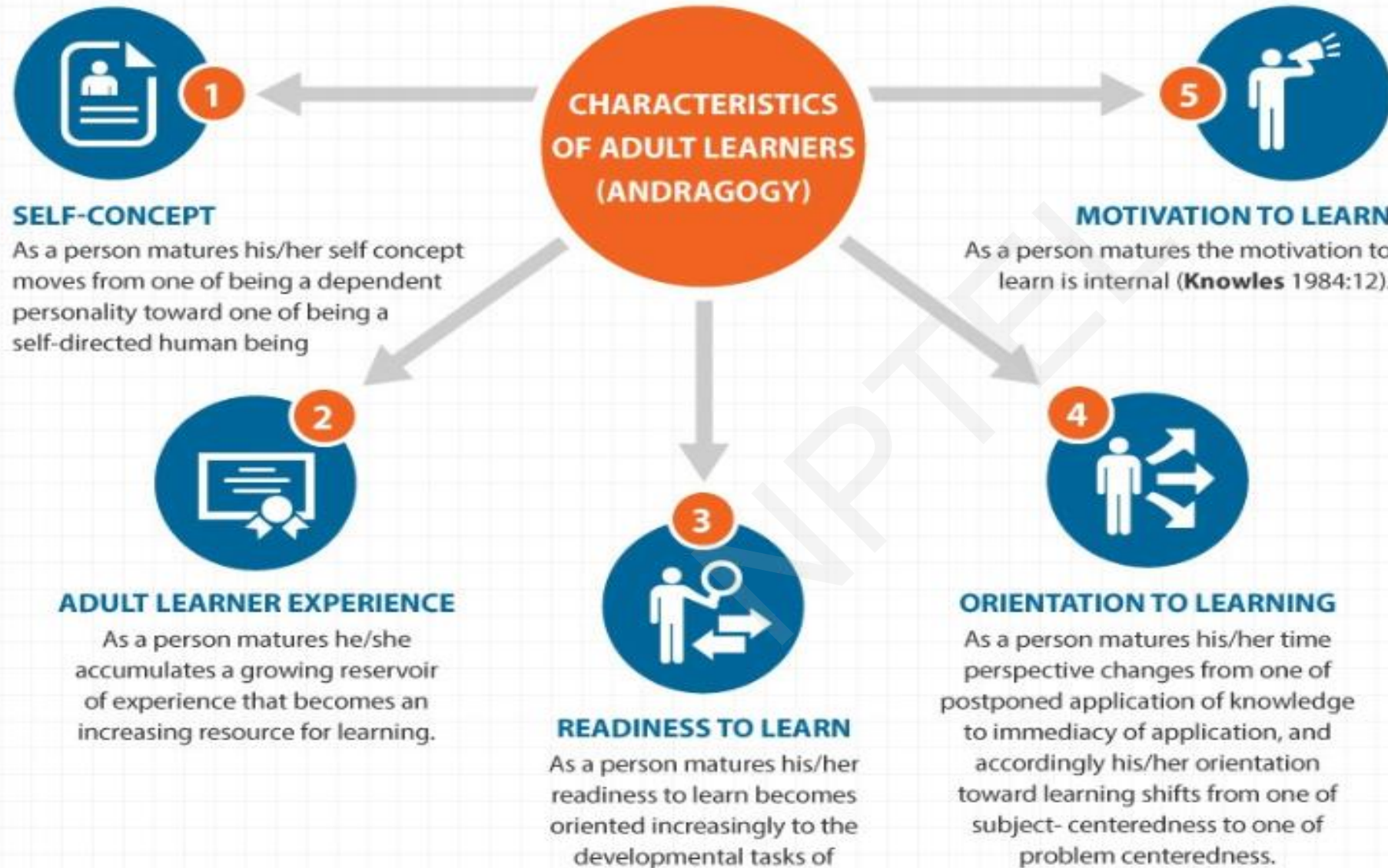
Psychology of E-Learning

PROF. ATASI MOHANTY
INDIAN INSTITUTE OF TECHNOLOGY
KHARAGPUR

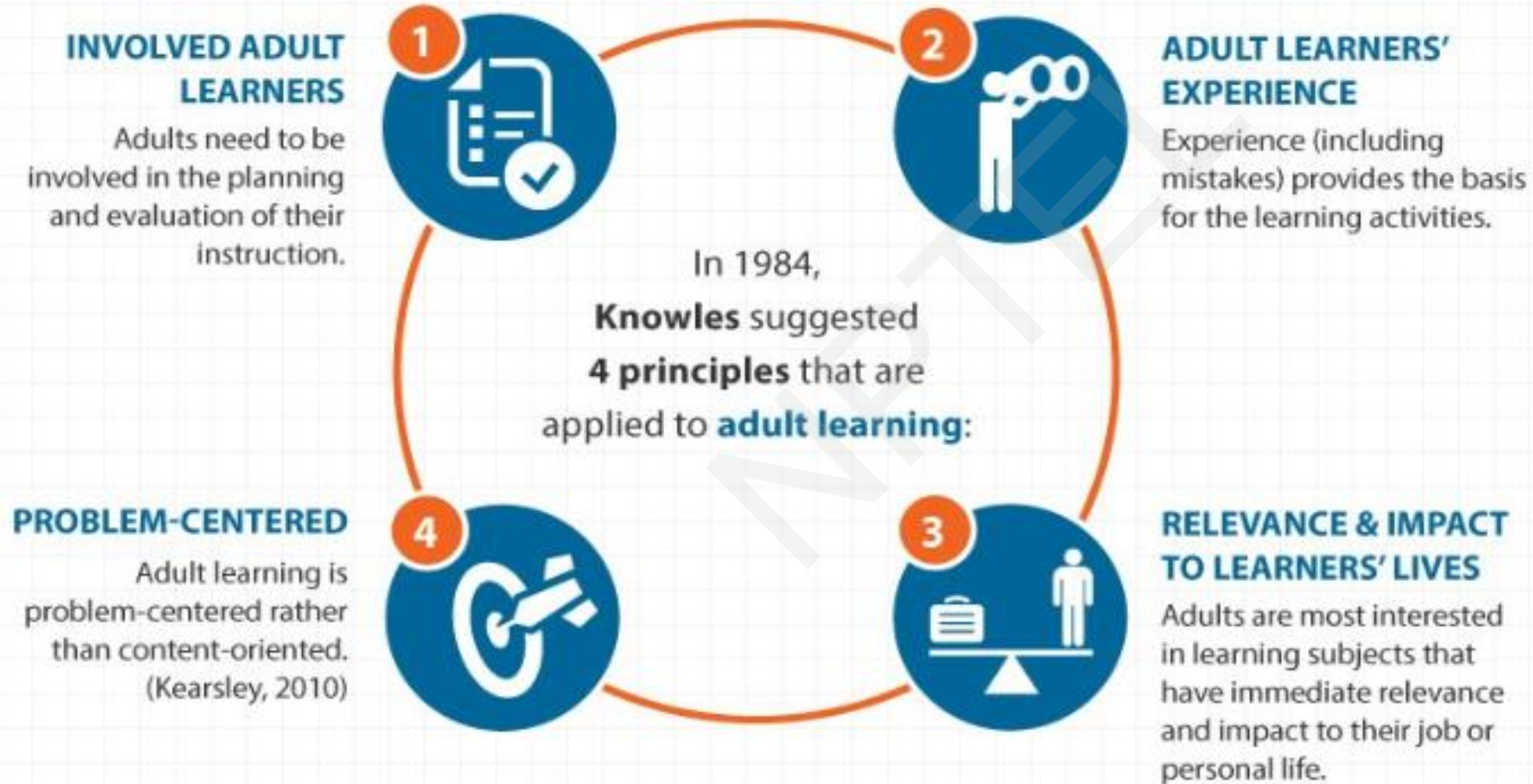


LECTURE : 45

In 1980, **Knowles** made 4 **assumptions** about the **characteristics of adult learners (andragogy)** that are different from the assumptions about the characteristics of child learners (pedagogy). In 1984, **Knowles** added the 5th assumption.



4 PRINCIPLES OF ANDRAGOGY



- **Principles For Designing eLearning To Engage Employees-**
- **Make Sure The Learning Is Relevant To The Learner**
- To **ensure engagement** is to make sure that the **learning is relevant** to the learner.
- Designing **content** and **activities** that are **directly related** to their **job role** or goals.
- **Make It Intrinsic**
- Intrinsic motivation is the desire **to learn for learning's sake**.
- It's what drives people **to seek out new knowledge** and **skills** even when there isn't an external reward or incentive involved.
- In eLearning, look for ways to **make the learning** itself **intrinsically motivating**.
- Making the **content interesting** and **engaging**, providing opportunities for **exploration** and **discovery**, and giving learners a **sense of autonomy** and control over their learning experience.





- **Keep It Satisfying**
- People are more likely to be engaged with **learning experiences that are satisfying** and enjoyable.
- One way to increase satisfaction is to provide learners with **opportunities to give feedback** and see their **progress over time**.
- To design **activities that are challenging but not too difficult** so that learners **feel a sense of achievement** when they complete them.
- **Make It Social**
- Social learning is another important factor in engagement.
- When learners are able **to interact with each other and learn together**, they're more **likely to be engaged with the content**.
- There are a few different ways to make eLearning social.
- One is to provide **opportunities for learners to collaborate on projects** or activities.





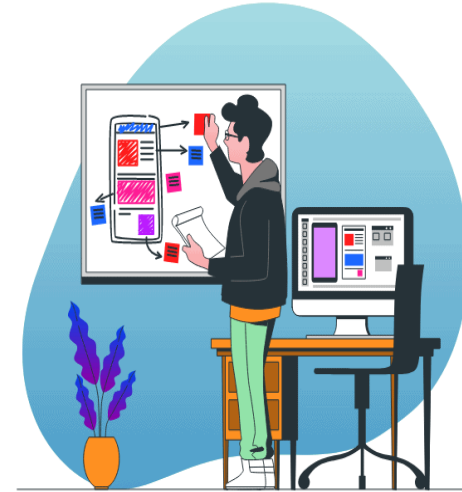
- To use **social media platforms** like **discussion forums** and **blogs** to encourage **interaction** and **dialogue**.
- It's also critical to **recognize employees** when they achieve a certain level in their learning.
- Ensure that **employees associate positive emotions** with **learning**.
- **Make It Personalized**
- Personalization is another key ingredient in engagement.
- When **learners feel like the content is tailored to their specific needs** and interests, they're more likely to be engaged with it.
- To **use adaptive learning technologies** that can adjust the **content based on each learner's individual needs**.
- To provide a **variety of content options** so that learners can choose what they want to learn about.
- To use learning platforms that automate this— make **personalized recommendations using AI tools** that help learners grow in necessary areas.





dlesStudio

Proof of Concepts and Prototypes in Product Design



- Cognitive Psychology in E-learning Design-
- It's important to look at our content much in the same way a cognitive psychologist would, so that we would **present knowledge in a format the human mind can quickly and easily understand**.
- The **basic units of knowledge** are commonly referred to as **concepts, prototypes, and schemas**.
- **Concepts** are large categories of knowledge in which similar ideas are grouped together.
- They can be **concrete** (like a book) or more **abstract** (like anger).
- **Prototypes** are the most recognizable examples of a concept.
- So, for **example**, when we think of a **computer**, a **MacBook** is the first thing that comes to mind, that's our **prototype**.
- **Schemas are the frameworks** we employ to understand the world.



- When one encounters information that doesn't fit with an existing schema, learning happens—either through **assimilation or accommodation**.
- **Assimilation** means broadening an existing schema, while **accommodation** means creating a completely new one.
- **For example, when you learn a new way to tie your shoes, you're creating a new schema.**
- **Information Processing-**
- We need to understand how these different units are processed.
- There are **two key principles of information processing**:
- **Data is gathered from our surroundings constantly and by multiple different systems.**
- Our information processing systems modify the data we collect.
- **We have to be mindful of how we present new information.**
- Since our processing systems tend to modify incoming information, **concepts need to be presented in multiple ways to drive the lesson home.**
- **Example-in public speaking, "Tell them what you're going to say, say it, and then tell them what you told them."**

Psychology of E-Learning





- The **format of learning** (be it classroom, computer-based, or blended) **impacts the data collection** as well.
- Think about what **distractions in the learning environment** may lure our learner's attention away.
- **How to make the mode of presentation more effective?**
- **Are there instructions and best practices we need to send to instructors along with the training materials?**
- **Knowing our Processing Limits**
 - We need to be mindful of the human mind's limitations of information processing.
 - One of the cornerstones of Cognitivism is its **focus on learning through self-discovery**.
 - e-Learning course developers **shouldn't over-weight their curriculums with content** that "tells learners";
 - Content should be presented in a way that **learners are forced to discover some aspects of the syllabus for themselves**.





- Presenting core concepts of a lesson in the form of high-level theories and learning points.
- Supplemental content that offers learners the opportunity to research and explore other elements on their own must be provided.
- This will allow them to form their own opinions about those core concepts laid out by the instructor.
- E-Learning is sometimes synonymous with "compressed learning".
- To deliver a set of slide presentations or videos, followed by online quizzes.
- That way, one can quickly breeze through a course in record time.
- A very crucial element of cognitive learning—one that's often touted as the biggest benefit of in-class teaching—interaction!
- Behavioral theorists hold that learners often learn more from each other than they do from an instructor rhyming-off bullet points.
- To enforce interactivity amongst online learners is to mandate group discussions and chat group participation as part of the overall course evaluation.





- As online learners often study at their own pace, it can add an element of boredom to the learning process.
- When learners aren't motivated to learn, they tend not to use their cognitive abilities to the fullest.
- The use of **simulations, role plays, and multi-player games** are all excellent ways **to foster interactivity among learners** and challenge them **to use** more of their **cognitive functions to learn new ideas** and difficult concepts.
- **Making Cognitive Learning Relevant-**
 - To give individual learners the ability to configure personalized learning paths.
 - Cognitivism recognizes that not all learners have the same level of cognitive abilities.
 - Hence, not every distance learner will progress at the same level; eLearning course designers should, therefore:





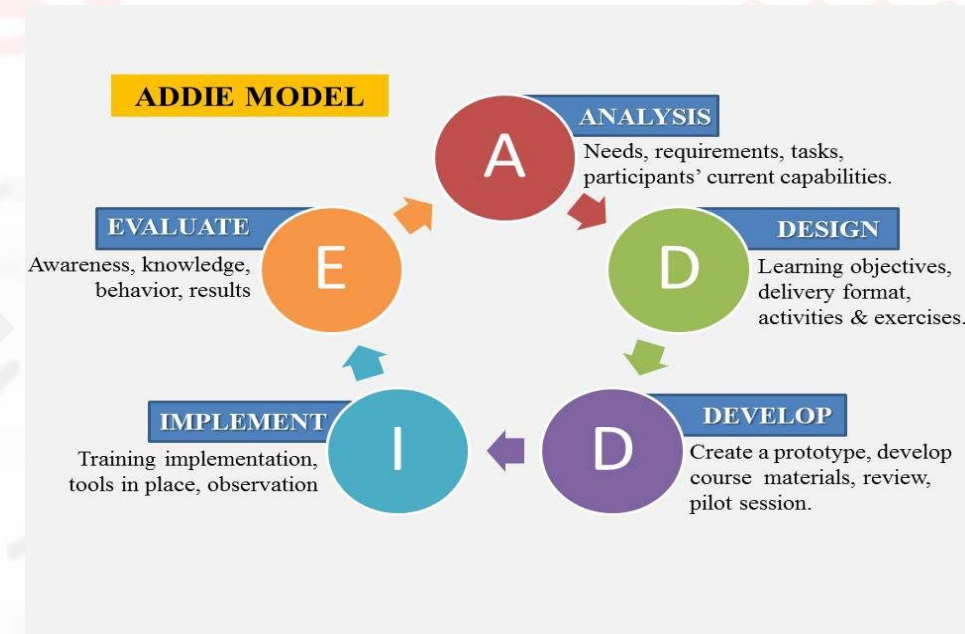
- Allow individual learners to sequence learning content the way they feel best fits their need.
- Build courses so that learners can skip past certain lessons, or repeat others multiple times, based on their individual comfort with the subject matter.
- Cognitive learning can be made more relevant in eLearning by allowing learners access to different mediums of learning.
- Should provide the main content in bullet points; to include additional clarification notes or annotations in the form of text, and append optional content (videos, audio files, and open-source content links) for individual learners to explore a subject further, should they wish to do so.
- Since Cognitivism deals with how individuals use their brains to "think" about what they are learning, eLearning content developers must do everything possible to challenge learners' minds to the fullest.
- To leverage the science behind cognitive learning, or "intelligent learning", as it is sometimes called, and make it relevant in today's eLearning environment.



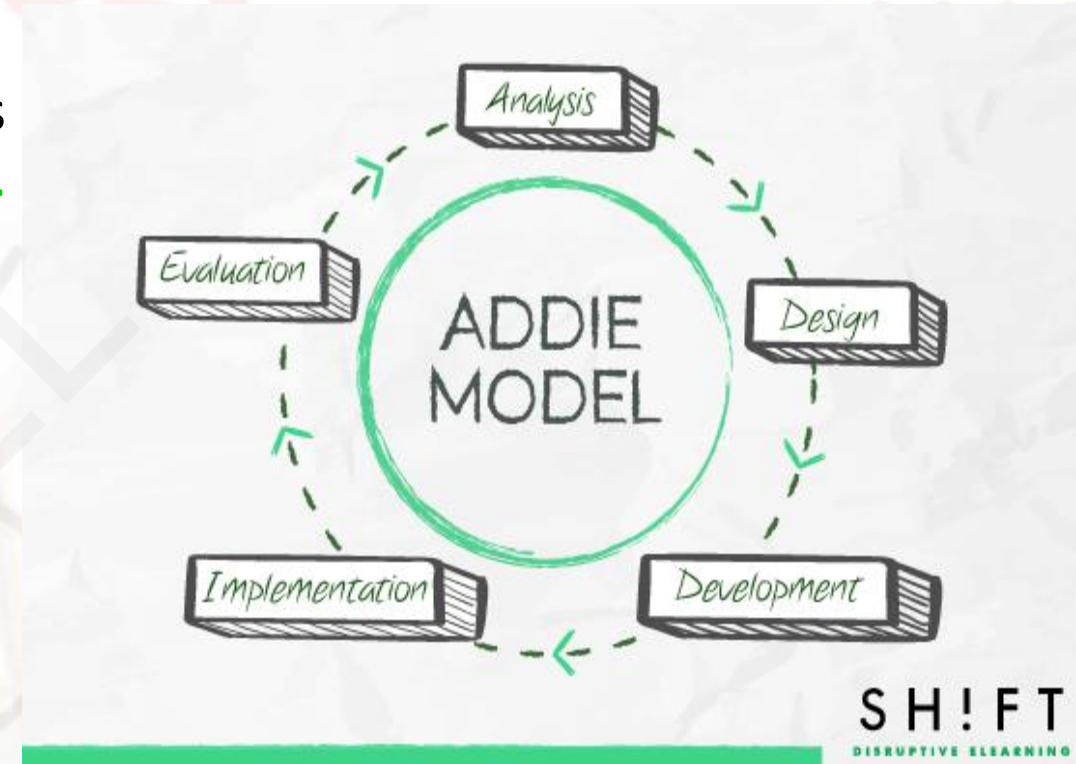
- **Instructional Design Models for E-Learning-**

- An instructional design model is used to define the **activities** that will guide the development **of eLearning projects**.
- It allows us to communicate the **purpose** and **reason behind a strategy**.
- A framework gives us the birds-eye view of all the major components that have to be included in the course.
- ADDIE Model-ADDIE stands for Analysis, Design, Develop, Implement, and Evaluate.
- **Each phase** of the model offers an **opportunity for iterations** and changes before moving to the next one. **Steps of the ADDIE process:**

- **Step #1 Analysis — Why is the training needed?**
- The instructional designers (IDs) answer this question after exhaustively collecting information and profiling target learners, and understanding the needs and expectations of the organization.
- **Analysis drives design** and the **development process**



- **Step #2 Design** — In this phase, IDs select the instructional strategy to follow, write objectives, choose appropriate media and delivery methods.
- **Step #3 Development** — IDs utilize agreed expectations from the Design phase to develop the course materials.
- **Step #4 Implementation** — The course is released/rolled-out, delivered, to the learners, and its impact is monitored.
- **Step # 5 Evaluation** — Is the course providing the expected results?
- IDs collaborate with the client and evaluate the impact of the course based on learner feedback, surveys, and even analytics.
- Once the evaluation is complete, the results are converted into actionable improvements.
- <https://www.youtube.com/watch?v=vSLv2KDYz9A>





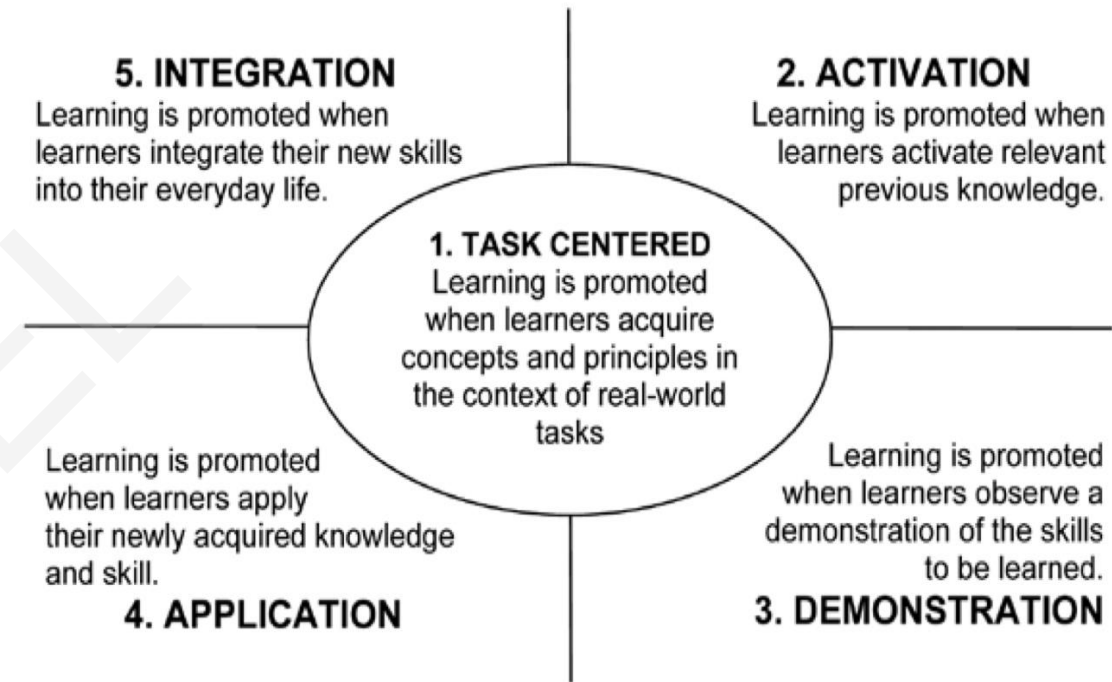
- **Merrill's Principles of Instruction (MPI)-**

- Proposed by **David Merrill** in 2002, this framework holistically integrates **five principles of learning**, namely:

- **Task-centered principle**
- **Activation principle**
- **Demonstration principle**
- **Application principle**
- **Integration principle**

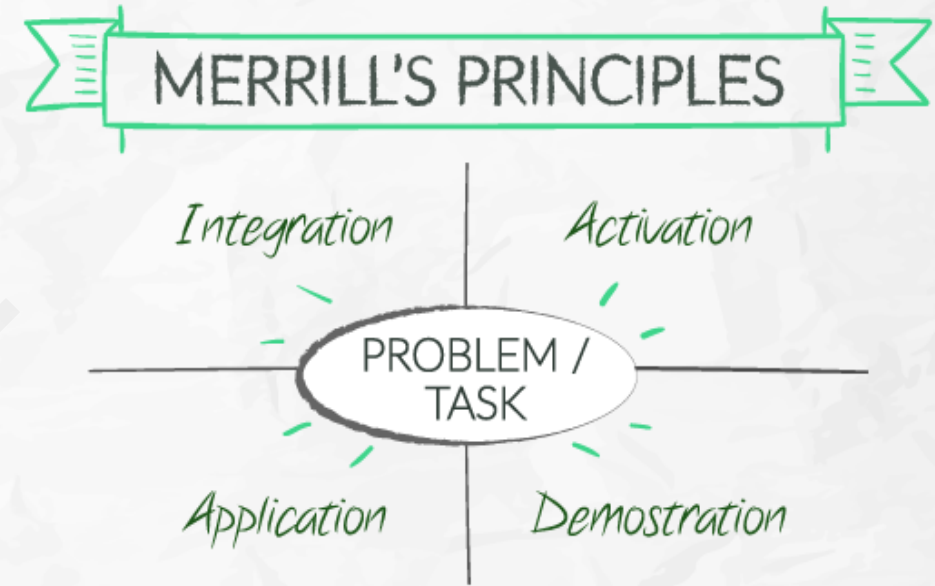
- These principles promote learning in the following manner:

- Learning starts with real-world problems.
- Students should be **able to relate to problems and tasks they can handle**.
- A course must **activate existing knowledge base** of the learner; hence aiding them **connect previous knowledge with the new one**.





- A course must demonstrate the knowledge so that it leverages different regions of the brain, hence retaining it longer.
- Allow them to apply new information on their own.
- Let the learners practice and learn from their mistakes.
- Let them see how the new material works in concrete situations.
- Must offer possibilities for integrating the knowledge into the learner's world through discussion, reflection, and/or presentation of new knowledge.
- **Gagne's Nine Events of Instruction-**
- **Robert Gagne** proposed a framework comprising of a series of events based on the behaviorist approach to learning.
- These events follow a systematic instructional design process, creating a flexible model where events can be adapted to cater to different learning situations.



SHIFT
DISRUPTIVE E-LEARNING

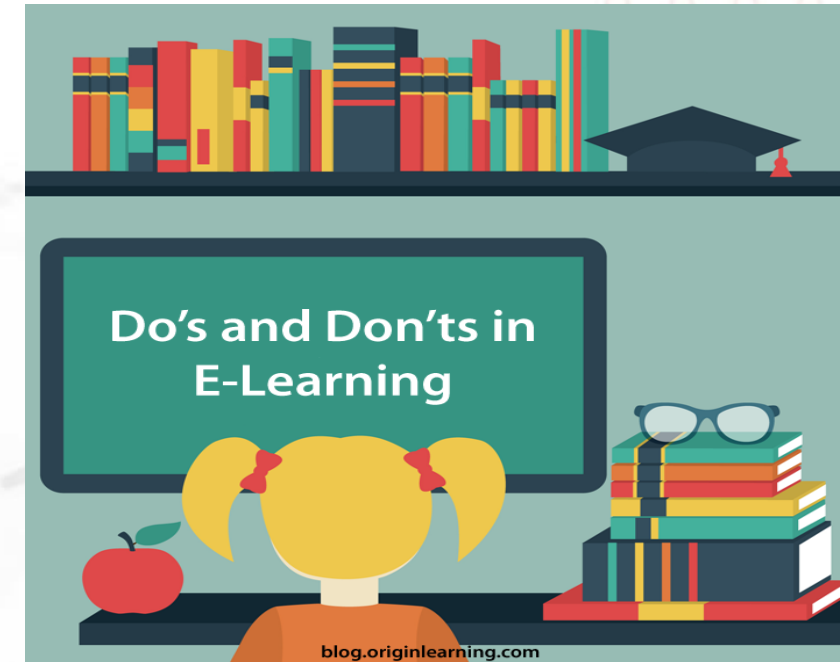


- It provides a sound structure for developing effective eLearning;
- **The Nine steps are:**
- 1. **Gain attention of the students** —with stimuli that catch and engage their brain (novel ideas or thought-provoking question, etc.)
- 2. **Inform students of the objectives** — Establish the expected outcomes and criteria for measuring achievement.
- 3. **Simulate recall of prior learning** —Leverage existing knowledge before introducing new knowledge and build on it.
- 4. **Present the content** — Deliver the content in easily consumable chunks.
- 5. **Provide learner guidance** — Guide them with examples, case studies, and other instructional support to supplement the content.
- 6. **Elicit performance** — Engage them with different activities that recall, utilize, and evaluate knowledge.

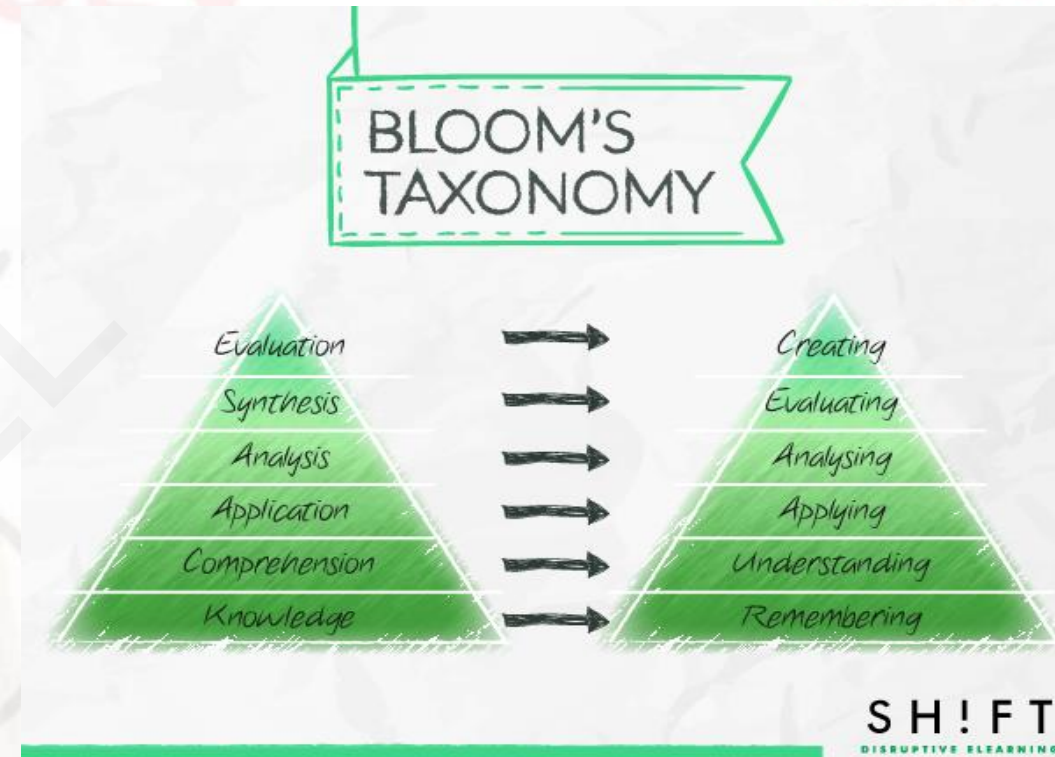




- 7. **Provide feedback** — Reinforce knowledge with immediate feedback (informative, remedial, corrective, etc.)
- 8. **Assess performance** — Test their knowledge with established (and transparent) criteria.
- 9. **Enhance retention and transfer to the job** — Use content retention strategies (concept maps, rephrasing, summarizing, job aids, etc.)
- <https://blog.originlearning.com/robert-gagnes-nine-steps-of-instruction-dos-and-donts-in-e-learning/>
- **Bloom's Taxonomy** –
 - In 1956, **Benjamin Bloom** created a classification system of measurable verbs to describe and organize the different levels of cognitive learning.
 - In 2001, the six dimensions were modified by **Anderson** and **Krathwohl** and are known as the "**Revised Taxonomy**."
 - <https://sites.google.com/apu.edu/tutoring/writing-resources>



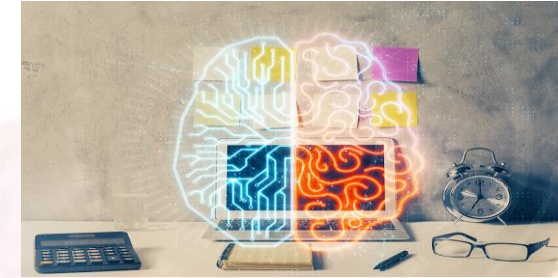
- ✓ The graphic here shows the **original (left)** and **revised (right) Taxonomy**.
- ✓ As given here, there's a new category at the top (Creating), three categories were renamed, and the categories are expressed as verbs instead than nouns.
- ✓ The taxonomy **pushes the learners** past the lower steps of learning (of knowledge and remembering) and **into the domain of deeper understanding, reflection, and application of knowledge** to develop a learner's individual process of solving problems.
- ✓ It is a great way for establishing **learning objectives that engage learners with the content** and **ingraining new knowledge and concepts**.





- **How People Learn Online-**

- eLearning allows individuals to access educational materials, participate in courses, and interact with instructors and other learners from virtually anywhere with an internet connection.
- Key characteristics of eLearning include:
 - **Online Delivery**
 - eLearning courses and materials are accessible via the internet, making them available 24/7, allowing learners to study at their own pace and convenience.
 - **Multimedia Content**
 - eLearning often incorporates a variety of multimedia elements, such as videos, audio clips, interactive simulations, animations, and graphics to enhance engagement and understanding.
 - **Interactivity**
 - Many eLearning platforms offer interactive features, such as quizzes, discussions, and assessments, allowing learners to actively engage with the content and receive feedback.





- **Self-Paced Learning**
- Learners can progress through the materials at their own speed, allowing for individualized learning experiences.
- **Flexibility**
- eLearning provides flexibility in terms of time and location, allowing learners to study from home, work, or on the go, making it particularly convenient for busy individuals and those with geographical constraints.
- **Accessibility**
- eLearning can be accessed by a wide range of individuals, including those with disabilities, provided the content is designed to be accessible and inclusive.
- **Remote Learning**
- With eLearning, physical barriers are removed, enabling people from different locations and backgrounds to access the same educational resources.





- **eLearning can take various forms**, including:
- **Online Courses**
- **Full courses** offered by educational institutions or eLearning platforms, covering various subjects and levels of complexity.
- **Webinars and Virtual Workshops**
- **Live, interactive sessions** conducted over the internet, often led by experts or educators.
- **Micro-learning**
- **Short, bite-sized learning modules** that focus on specific topics, suitable for quick consumption and on-the-go learning.
- **Blended Learning**
- A **combination of traditional in-person instruction and eLearning components**, offering a more flexible and varied learning experience.



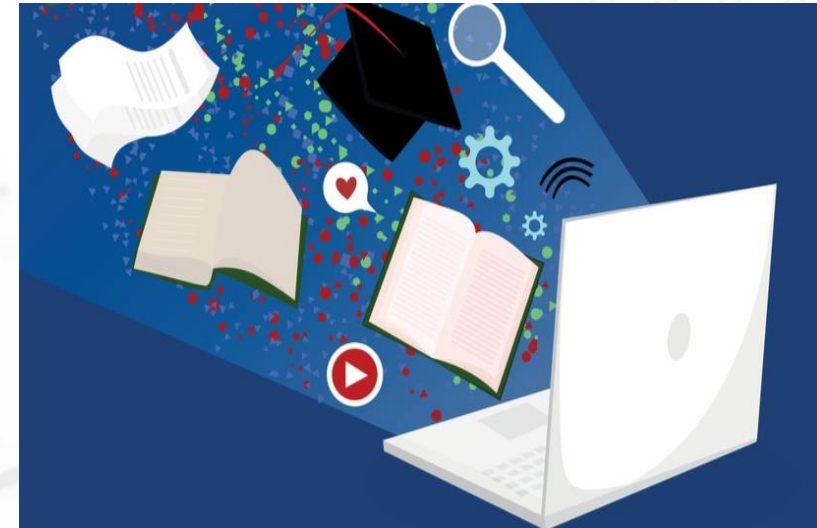


- **Massive Open Online Courses (MOOCs)**
- **Online courses** open to a large number of participants from around the world, often offered by prestigious universities and institutions.
- eLearning has gained significant popularity in recent years due to its **accessibility, convenience, and ability to cater to diverse learning needs**.
- It is used in various fields, including **formal education, professional development, corporate training, and lifelong learning** initiatives.
- Understanding **how individuals process information, engage with content, and retain knowledge in a digital environment** can help **educators create more engaging and successful eLearning courses**.
- **Key psychological principles and factors :**
- **Cognitive Load Theory**
- Cognitive load refers to the **mental effort required to process information**.
- In eLearning, it's essential **to manage cognitive load** to prevent overwhelming learners.





- Effective eLearning courses often break down complex topics into smaller, manageable chunks, use multimedia elements strategically, and provide opportunities for spaced repetition to enhance learning and retention.
- **Multimedia Learning**
 - Leveraging multimedia, such as **videos, images, animations**, and **audio**, can enhance learning in eLearning environments.
 - According to the Multimedia Learning Theory by **Richard Mayer**, **combining visual and auditory information** can **improve understanding** and **retention**, provided it is presented coherently and not excessively.
- **Self-Directed Learning**
 - Online learning environments often require **learners to take more responsibility** for their learning process.
 - **Autonomy** in learning can be empowering, but it also demands **self-discipline** and **self-regulation**.
 - Designing eLearning courses that promote self-directed learning and provide opportunities for **learners to set goals** and **monitor their progress** can lead to **better outcomes**.



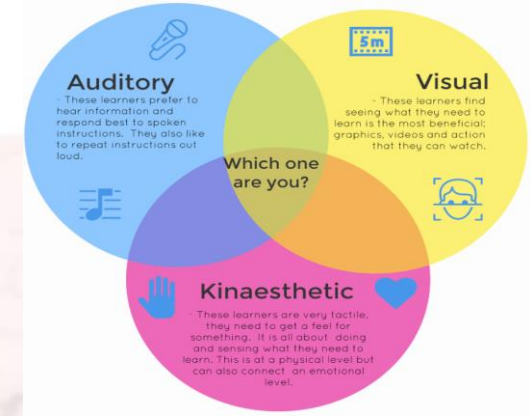


- **Social Learning**
- Although eLearning is often a solitary activity, incorporating social elements can boost engagement and learning outcomes.
- Discussion forums, group projects, and collaborative activities can foster social interaction and knowledge sharing among learners.
- **Gamification**
- Gamification elements, such as badges, points, leaderboards, and challenges, can enhance motivation and engagement in eLearning.
- These game-like elements tap into people's intrinsic motivation and desire for achievement.
- **Feedback and Assessment**
- Constructive feedback is crucial in the eLearning environment.
- Immediate and informative feedback helps learners understand their progress and areas for improvement.
- Frequent formative assessments, quizzes, and interactive exercises can also reinforce learning.





- **Learning Styles and Preferences**
- People have diverse learning styles and preferences. Some learners may prefer **visual content**, while others may favor **textual explanations**.
- **Designing** eLearning courses **with varied formats** can accommodate different learning styles and enhance engagement.
- **Attention and Focus**
- Online environments can be distracting, so it's vital to **design** eLearning courses that **capture and maintain learners' attention**.
- **Clear organization, interactive elements, and visually appealing content** can help sustain focus.
- **Retention and Retrieval**
- The **spacing effect** and **retrieval practice** are valuable techniques to improve long-term retention.
- Spacing out learning sessions over time and incorporating **quizzes** or **review activities** can aid memory **retrieval** and **reinforce learning**.





- **Motivation and Incentives**
- **Intrinsic motivation**, driven by interest and curiosity, tends to result in more enduring learning outcomes.
- **Extrinsic motivation**, such as certificates or career advancement opportunities, can also play a role in encouraging participation.
- **Emotional Design**
- Emotions can influence learning and memory.
- Creating a **positive** and **supportive learning environment** can **increase learner engagement** and **facilitate** the integration of **new knowledge**.
- **Accessibility and Inclusivity**
- Consider learners with different abilities and ensure that eLearning materials are accessible to everyone.
- Providing **closed captions**, **alternative text for images**, and other **accessibility features** can make **learning** more **inclusive**.



Inclusive Design in an Online Learning Environment



How might we encourage faculty to adopt inclusive teaching practices?

- **Instructional Design Strategies For Rapid eLearning –**
- **Strategies For Immersive eLearning-**
- **Scenarios**
- Learning happens best when it provides **proper context** and encourages learners to take action.
- Scenario-based eLearning provides the said context to learners in the form of **simulated job-related situations** where learners have to judge a situation, use the information they have at hand, and make the best possible choice to solve the problem.
- It helps learners with **situational awareness, critical thinking,** and **decision making**.
- Scenarios are very useful to teach topics where there are no right or wrong actions.
- They also help bring about **behavioral changes in learners**.
Soft-skill and **compliance training** are good examples of where scenarios can be used as an instructional strategy.



- **Guided eLearning**

- The strategy of guided learning uses a ‘coach/ mentor’ along with voiceovers.
- This coach/ mentor (who can be a senior, a manager, or anyone that learners can relate to in the job context) essentially performs the role of a virtual instructor and guides the learners through the course.
- **Learning Through Exploration And Discovery (LEAD)**
- This learning strategy uses a learning ‘journey’ map to give learners the freedom to explore the course at their own pace.
- Images and/or icons are used on the map as visual indicators to show learners their progress and the destination ahead.
- It is a very effective strategy for on boarding training where learners need to know a variety of information and can afford to choose their own learning path.



HUMBOLDT
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GUIDED E-LEARNING

- flexibles und ortsunabhängiges Lernen
- monatlich 5 inkludierte Lernstunden
- persönlicher Coach
- speziell für Jugendliche



• Simulations

- If scenarios give learners a **context similar to their job** environment, simulations mimic them in a **virtual setting**.
- This high-level learning strategy helps learners gain **hands-on experience** in a risk-free environment.
- Most **software training is simulation-based**.
- Authoring tools help develop **simulations in 3 phases** – Watch, Try, Do.
- In the '**Watch**' phase, learners are shown how to execute different tasks in a software with instruction callouts..
- In the '**Try**' phase, learners execute the steps on their own, aided by prompts and hints.
- In the '**Do**' phase, learners execute the steps without any guidance.

• Storytelling

- Storytelling is one of the most common instructional strategies used in eLearning.



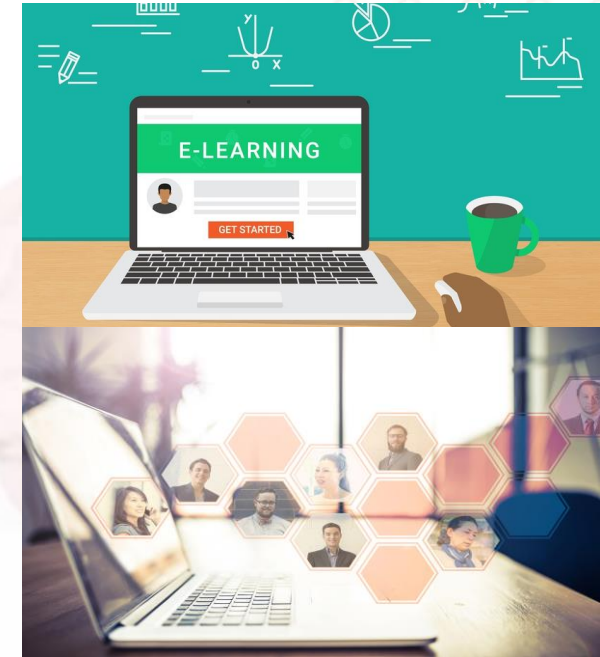
- The essence of storytelling lies in its **ability to connect with learners emotionally, build empathy**, and make even the most boring training subject interesting.
- **Case studies** and **comic strips** are the best examples of storytelling in eLearning.
- **Gamification**
- Gamification is another very popular instructional strategy used in eLearning courses.
- It involves the **use of game elements** – **points, levels, rewards, timers** – to **facilitate effective knowledge transfer**, build an emotional connection, elicit responses from learners, foster competition, and help retain knowledge longer.
- It is a popular **tool of edutainment** where **fun** and **learning** are provided together for a more engaging learning experience.





• An Inclusive eLearning Content Design-

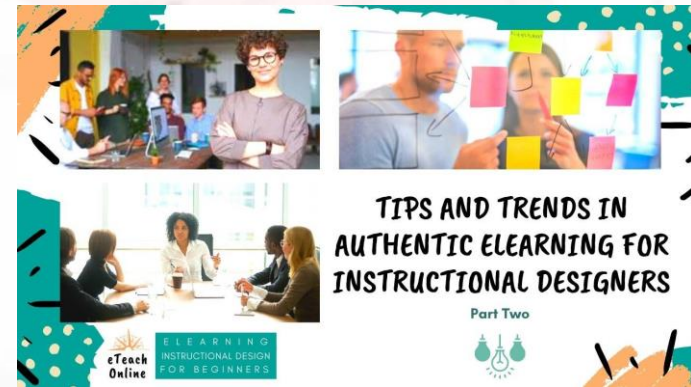
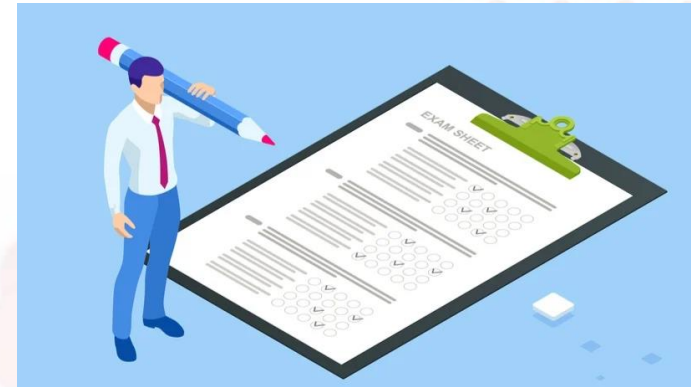
- Each learner comes to y/our training with different backgrounds and expectations.
- **Inclusive design** is not a design practice, but **a philosophy** of designing **for one** and extending **to many**.
- It's about **creating the best possible experience** that everyone can participate in.
- **Dialogue** Between The **Learner** And The **eLearning Design**
- Since dialogue is one of the most powerful instruments in learning, we include it in the eLearning design.
- This can be done by **using inclusive pronouns** ("what do you think?"), **spoken language expressions** ("that's right!") or simply **telling the story** from the learner's point of view ("imagine you are...").





- **Explorations In Authentic Environments**

- Involving the learner in the situation through language and authentic landscapes make for compelling learning experiences.
- It's The Way You Make Them Feel
- It's not what you say, but how you say it that matters.
- People remember how the messages make them feel.
- Do we hide behind scientific objectivity or come close to the world of experiences of the learner with spoken narratives?
- Production Instead of Recognition
- Best eLearning experiences offer us an opportunity to create our own learning products.
- Instead of surfing the content, we can have the learner dive deeper to produce their own videos, images, graphs, tables, or other manifestations of their understanding of the material.
- Production usually leads to effective learning.





Psychology

- **Personal Choices**
- We have our preferences and dislikes.
- Learning design should tap into this.
- Whenever the learner's personal interests and ideas coincide with the learning content, there's a bigger likelihood for design success.
- My Kind Of Guide For A Learning Design Experience
- Learning alone is hard.
- That's why it's important to interact with the learner in a personal, emotional, and inclusive manner.
- We want to be their coaches, their guides, their travel partners.
- We want to take them on a journey where they are allowed to feel, explore, and make errors in a safe environment.
- This is where language comes into play, as do the words we choose that can either include or exclude.
- We want our **learners to feel empowered, exhilarated**, and, in the end, **confident**.





- Improving the learning experience in **eLearning platform begins with understanding the learner.**
- Learners' need assessment questions:
 - What does your student want to accomplish or learn?
 - What is the student's level in regard to previous knowledge, abilities, and behaviors?
 - What does the student need to be able to know, do, or change after completing the course?
 - What tools does the student need to reach their goals?
 - What metrics will be used to measure your course's impact and students' success?
- With a better understanding of our students, we can take on the **process of constructing a valuable learning experience** that is both pleasant to take part in and helps them reach their **desired outcomes.**





THANK YOU

